Design of a High Rise Tower

7KS2M0 - Design Project High Rise Building

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Abstract

This report presents the integrated architectural and structural design of a sustainable high-rise tower on the VDMA terrain in Eindhoven. The project aims to minimize the carbon footprint of material use while maintaining structural integrity, flexibility, and future adaptability. After evaluating three design variants, a stepped timber tower with CLT cores and a timber truss stability system was selected based on its low embodied carbon and contextual fit. Extensive calculations were conducted to optimize dimensions, assess structural performance, and ensure compliance with Eurocode safety requirements. Fire safety, acoustics, and maintenance were critically addressed, and alternatives were proposed to resolve identified durability issues. The resulting design demonstrates the feasibility and limitations of pursuing low-carbon timber high-rise construction within current regulatory and material constraints.

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1 Introduction

1.1 Elaboration of the project

As global populations continue to concentrate in cities, space becomes a critical limitation in urban design. These high population densities prompt developers and urban planners to build vertically, enabling cities to accommodate a larger number of people and businesses within a smaller area. Besides capacity, high-rises are also seen as an indication of economic prosperity and modernity, which further motivates their construction in cities.

Currently, this trend is not just about building taller, but also smarter and more sustainably, as the environmental impact of high-rise is well known. There's a growing need for green building technologies, energy efficiency, and smart infrastructure within these towers to address environmental concerns. Furthermore, advancements in construction materials and engineering enable increasingly complex designs, pushing the boundaries of height and architectural innovation.

Designing high-rise buildings presents significant challenges, primarily due to the scale of the building and consequent complexity. Structural principles used for low-rise do not always apply, requiring innovative solutions to resist more significant wind loads, seismic forces, and the building's own weight. Vertical transportation, such as elevators, must be well-developed and ensure the safe travel of building users. Fire safety becomes a critical concern, with complicated evacuation strategies and suppression systems integrated into the building. Furthermore, achieving sustainable performance, optimizing natural light, and ensuring occupant comfort at extreme heights adds complexity.

This report describes the design process of a high-rise tower, designed for the VDMA terrain in Eindhoven. The building will be used as an office building, but is designed with the possibility of conversion to residential apartments in mind. The focus of the design process is on the structural design of the building, but architectural and building physical aspects are also considered to ensure an integrated design. A design goal is chosen to provide coherence in all decision-making in the project. This goal is considered during all stages of the design process.

The design process starts with the development of three variants, for which the architectural design is elaborated, as well as the main idea for the structural design. From these variants, one is chosen to continue the design process with. For this variant, two different stability principles are analyzed. Based on the optimal solution, the structural design is further developed. The design is optimized by analyzing potential risks that it may present, which are addressed using an integrative approach across multiple disciplines. The report concludes with an analysis of the design's feasibility and a propositions for further development.

1.2 Description location

The project site is situated on the VDMA terrain in central Eindhoven, between the Vestdijk and the Tramstraat, as shown in Figure 1. It is currently undergoing a transformation into a mixed-use urban hub. The industrial heritage is a focal point in these transformations. Current plans for the urban space include homes, cultural spaces, and stages, making it an ideal location for a high-rise office tower. Currently, the project site consists mainly of parking spaces. The VDMA building, which is also included within the project's bounds, is permitted to be demolished or utilized in the project. [1]

Figure 1: Project site

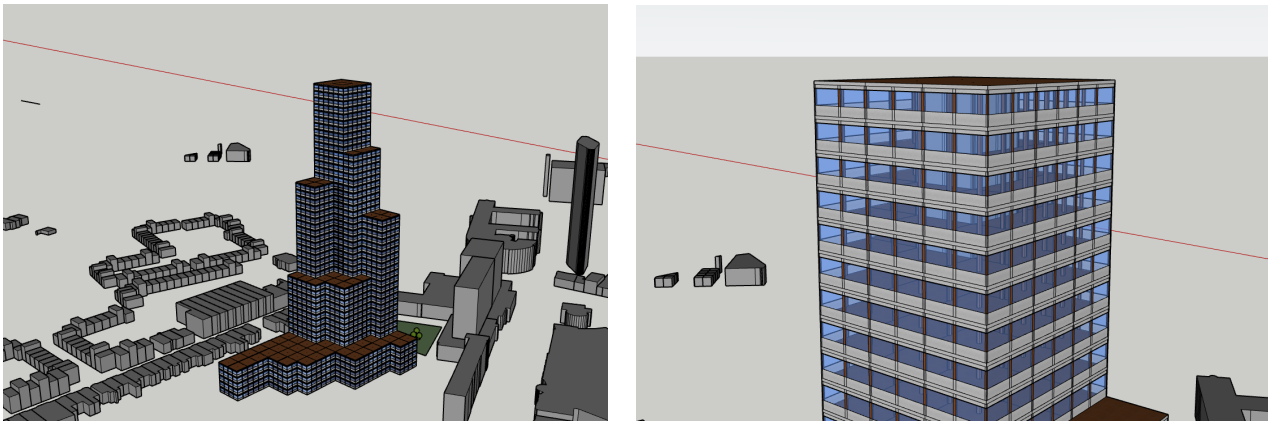
1.3 Design goal

To give the building design a clear direction, a design goal was chosen. This goal was consistently applied throughout the entire process and was taken into account in all decisions made. As high-rise structures are increasingly needed due to high housing demand, but these large structures can be detrimental to the environment, the goal of this assignment is to design a high-rise building with a low carbon footprint. To reduce complexity and remain within the scope of this project, only carbon emission from material use is considered. To assess all variants evenly throughout the project, all values used are found in the ICE embodied energy and embodied carbon database (V2.0) [2].

2 Design variations

2.1 Variation 1

The first variation features a timber column and beam structure with a concrete core. The columns and beam structure will result in an open structure, allowing for complete flexibility in the event of repurposing the building. The building will take a step-like shape to fit in with the dynamic view the surrounding neighborhood presents, with the highest tower located on the south side of the plot to minimize shadows on surrounding dwellings. However, the definitive shape of the building is not yet determined, as the main idea for this variation is to allow flexibility in this regard. The building will consist of multiple towers, all built up of a column and beam structure with the same dimensions. This will enable a parametric study of the tower's shape, examining variations with different tower heights, widths, and lengths. The combination of different towers will be a key factor in the stability design of the tower. Visualizations of this design variant are shown in Figure 2.

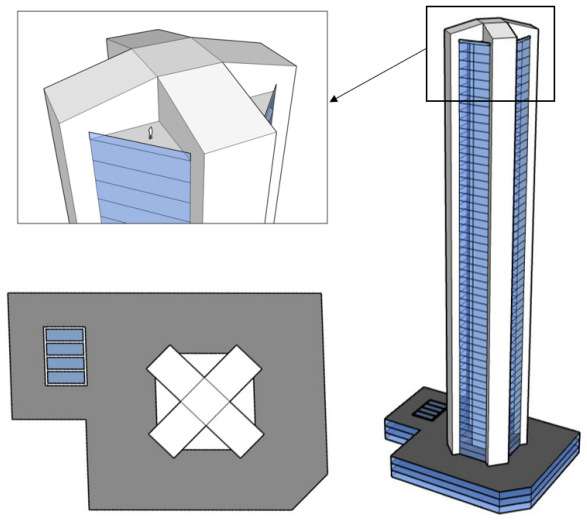


(a) (b)

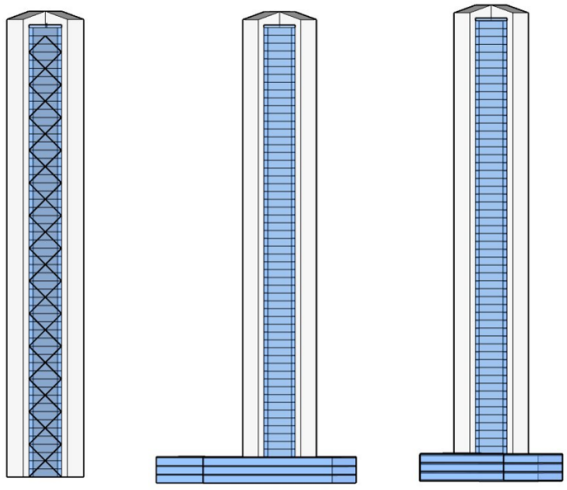
Figure 2: Variant 1 showing (a) the full building and (b) the open interior space due to the column-beam structure

2.2 Variation 2

The second design variant features a slender tower with four compartments positioned at its corners. Rising vertically from a broader base, the tower maintains a straight and elegant profile. The base serves as a social hub, intended for public use and interaction. Residential spaces are distributed along the vertical length of the tower. In Figure 3a, the tower is shown in a perspective view. The adjacent zoom-in highlights the top of the tower. The bottom left of Figure 3a presents a top view of the structure. A roof window is integrated into the design, allowing additional natural light to enter the building's base.



(a) Variant 2 showing (at the right) perspective view (left bottom) top view of the building



(b) Variant 2 showing (at the left) structure of the tower (at the right) elevations of the building

Figure 3: Variant 2 visualisations

In Figure 3b, the structural concept of the tower is illustrated at the left. The corner compartments function as cores or shear walls, acting as primary stability elements. These are positioned as far

outwards as possible to maximize the building's moment of inertia. The bracing between the cores ensures that they function together as a cohesive structural system. Across the remaining facade, windows are used extensively to allow ample daylight into the building. At the right of Figure 3b, the building's elevations are shown. As can be observed, the tower has a relatively simple form, yet the articulated corners introduce additional depth and complexity to the design.

2.3 Variation 3

This design variant features a timber column-and-beam structure combined with a central concrete core, offering both structural integrity and spatial flexibility. The use of large, open floor plates enables potential future repurposing of the building. The architectural concept envisions a slender, vertically stepped high-rise that integrates into the Eindhoven skyline. The stepped massing not only serves architectural purposes but also contributes to the structural stability by creating a tiered load distribution. Strategically positioned on the southern edge of the site, the tower minimizes shadow impact on the adjacent low-rise neighborhood. Extensive glass facades reinforce the visual lightness of the structure, though this design remains at the volumetric study stage. Visualizations of this design are shown in Figure 4.

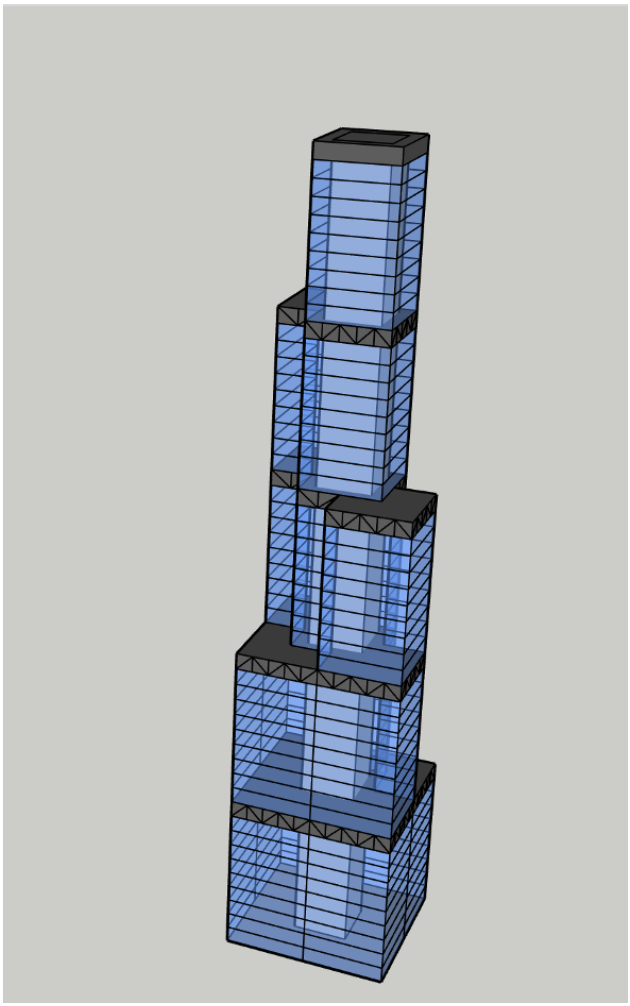


Figure 4: Variation 3

2.4 Comparison in relation to the design goal

To evaluate the three design variants in light of the project's design goal, each was assessed based on its material composition and potential for low embodied carbon. Variant 1 combines steel bracings, a concrete core, and CLT shear walls. While the inclusion of timber contributes to carbon reduction, the reliance on steel and concrete moderates its environmental performance. Variant 2 adopts a timber column-and-beam system with a concrete core and a straightforward volume. Its compact form and predominant use of timber result in the lowest embodied carbon among the options. Variant 3 also uses a timber column-and-beam structure with a concrete core but features a stepped form and extensive glazing. Although this increases the carbon footprint, it offers strong spatial flexibility and integrates well with the surrounding context.

2.5 Chosen variant

Considering the similarity between the first and third variants, and the outcome of the comparison in terms of the design goal, the step-design tower is chosen for the continuation of the project. It was also found to blend in best with the surrounding cityscape and allow for adaptability in the event of repurposing. In both variants, a timber column and beam structure is used with a concrete core for stability. However, in Section 3.3, other stability principles will be considered to ensure fulfillment of the design goals.

Neither one of the variants presented a set shape or design of the tower, as the main idea is to maintain flexibility in the design. To ensure that, regardless of the shape of the different towers, the building would remain within the allowed area. Figure 5 shows the build-up of the grasshopper model. All towers are adjustable in height, width, and depth. This allowed for easy exploration of different shapes.

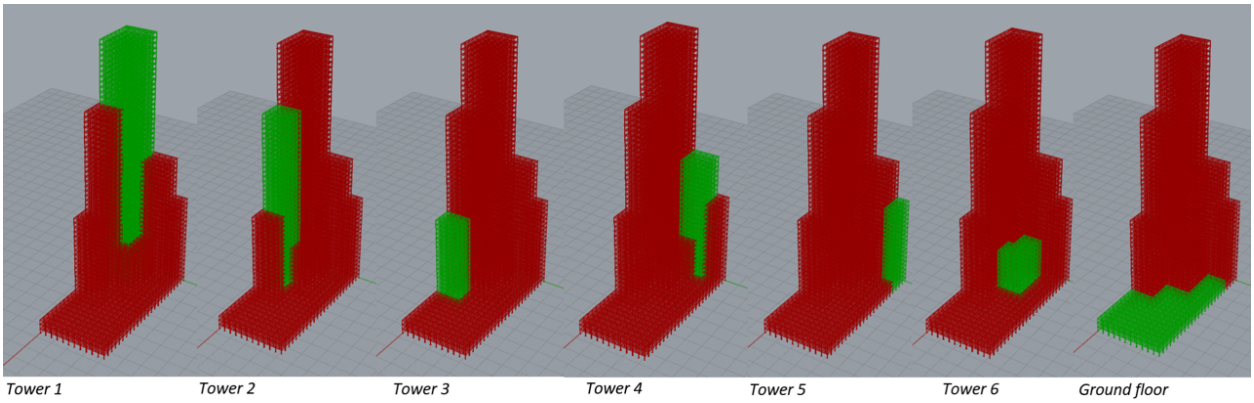


Figure 5: Elaboration grasshopper model

To accommodate the stepped massing concept and maintain adaptability in the design, the tower was subdivided into seven distinct towers. Each tower differs in width, length, and number of floors, allowing for a varied layout while fitting within the urban and environmental constraints. This design of the tower is determined to allow for the external stability diagonals, as elaborated in Section 3.3. This modular approach supports flexibility in both construction and potential future repurposing. The floors have a free height of 4 meters, excluding floor systems. Table 1 summarizes the key dimensions and floor distribution of each segment. In figure 6 a render can be found

Table 1: Overview of tower floor areas and dimensions

Floor area	Width [m]	Length [m]	Floors
A	30	30	38
B	15	30	18
C	15	30	14
D	30	15	18
E	30	30	10
F	15	30	6
G	30	30	1

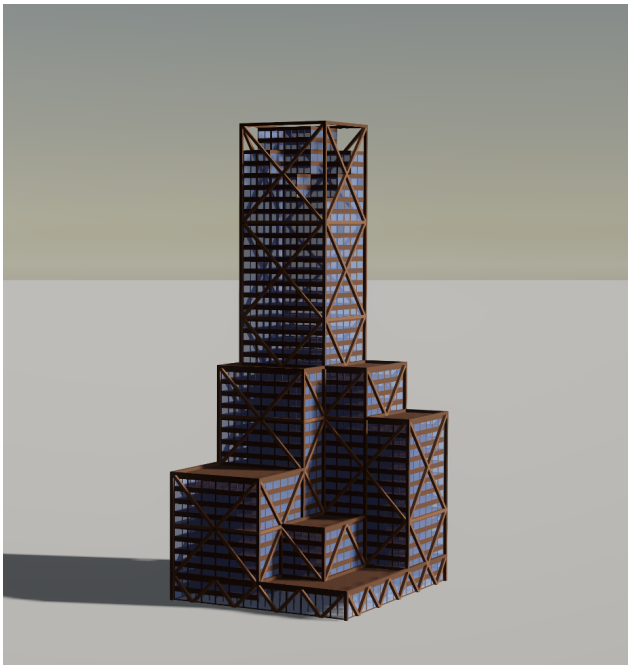


Figure 6: Render of chosen variant and final tower configuration

3 Stability principles

For the tower design, two structural stability principles are explored: a concrete core and a timber truss system. These two systems are initially assumed to be the most effective in meeting the project's design objectives. A comparative analysis will be performed to evaluate their performance, after which one system, or a combination of both, will be selected for the final design.

3.1 Concrete cores

The first proposed stability system is based on the use of concrete cores. Given the bundled configuration of the towers, placing a concrete core in each individual tower would ensure overall stability. However, due to the significant spacing between the towers, these cores would need to be interconnected to act as a unified system. One approach is to rigidly connect the cores, requiring the intermediate floors between towers to have a thickness comparable to that of the cores themselves (approximately 1 meter). This solution would consume a considerable amount of space, reduce the

functional floor height, and demand a large volume of material. As a result, it would be inefficient and conflict with the design goal of minimizing carbon emissions.

An alternative method involves connecting the cores using beams that are hinge-connected to each core. In this configuration, for example, with two adjacent towers of equal height, each core would carry half the load and function independently. Although this system provides stability, it does not fully utilize the potential for shared structural performance, which could otherwise reduce material use even further. Nevertheless, given the distance between the towers, the hinge connection with beams proves to be more efficient than the use of excessively thick floor slabs.

To evaluate the structural requirements, an Excel model (Appendix A.1) was used to determine the necessary core thickness for one of the towers. The highest tower, standing at 150 meters, was selected for this analysis. Inputs for the calculation included core dimensions, wind load, and concrete strength class. Using these parameters, the moment of inertia and the resulting displacement at the top of the core were calculated. Note that the concrete strength class is multiplied by 0.3 to indicate that the concrete is cracked. The core was modeled as a cantilever column fixed at the base, with the cross-section representing the entire core volume (Figure 7). Various dimensions were tested to identify an optimal balance between structural performance and spatial functionality, for instance, accommodating toilets, staircases, and elevators located within the core.

As a baseline, a 12 by 12-meter core was used for the calculations. This serves as a preliminary reference for later assessments of carbon emissions resulting from concrete production. The analysis indicated that a core thickness of 1.5 meters is required to achieve a top displacement below L/500, which was taken as the maximum allowable deflection. This limit is derived from an average of various design references and is applied here solely for initial calculations. Assumed is also that the deflection at the top of the building is defined in the dimensions of the core. The stresses in the structure are assumed to be significantly lower than the maximum stresses of the concrete. The maximum concrete class is chosen because of the assumption that the highest material strength with the smallest dimensions is equal to the lowest carbon emissions.

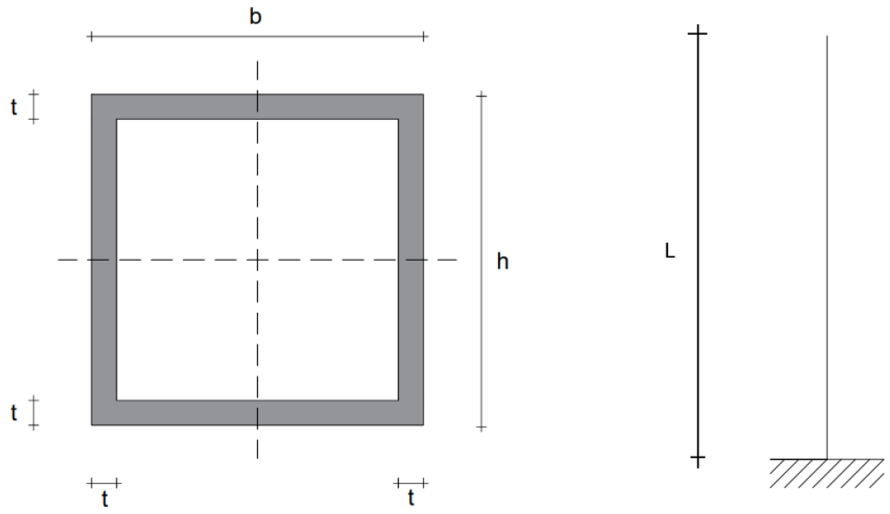


Figure 7: The cross section of the core and the mechanical system used for the core calculations

3.2 Timber truss-system

The second option for providing structural stability is a timber truss system, based on the concept illustrated in Figure 8. It is assumed that all connections within the truss are hinged. This system is further developed using an Excel model, detailed in Appendix A.2. All structural elements within the truss are made of timber.

The Excel sheet is used to explore different cross-sectional dimensions, aiming to limit the deflection at the top of the structure to L/500, consistent with previous calculations. The analysis considers both bending and racking shear: bending behavior is primarily influenced by the column dimensions, while racking shear is governed by the performance of the floors and diagonal members.

Key input parameters include the wind load, the modulus of elasticity of GL32h, and the building's geometric dimensions. GL32h is selected for the same reason as the use of a high-strength concrete class in the alternative system: to efficiently meet the structural demands.

The results indicate that the timber columns should be 1.5 × 1.5 meters in size, utilizing solid timber elements. The floor beams and diagonal members are each sized at 1.75 × 1.75 meters. However, it is worth noting that the current analysis is 2D, which means it does not fully capture the system's behavior in three dimensions. For instance, in the front view of the truss, multiple rows of columns may exist behind the visible plane, contributing to the structure's overall stability. This means that the individual column sizes used in the 2D model might be larger than what would be necessary in a full 3D configuration.

This interaction has not yet been considered, but it will be incorporated into the 3D structural analysis at a later stage.

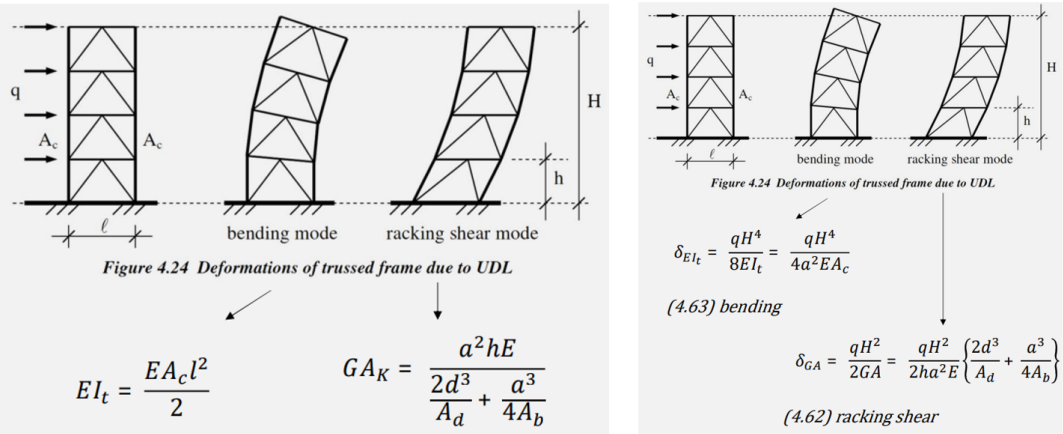


Figure 8: Timber truss stability calculation principle

3.3 Comparison in relation to the design goal

The decision between the two stability principles is made based on the design goal, which states that the CO_2 emission of the construction will be minimized. The CO_2 emissions of the specifically selected stability elements will be compared, as any other elements are expected to be similar, if

not the same. For the concrete cores, this means the cores themselves will be considered, as well as a concrete floor element every four stories (16m) to ensure diaphragm action and thus stability. For the calculation of the timber truss system, the timber diagonals are taken into account as well as the cores. Even though the cores are no longer structural, this provides a fairer comparison, as the cores will not entirely disappear when a timber truss structure is used.

Two variants are considered for the cores: non-loadbearing concrete and CLT cores. The calculation assumes three cores for all variants. The results are presented in Table 2. A detailed calculation is provided in Appendix A.3.

Table 2: Comparison CO_2 emission different stability systems

Stability system		**Total CO_2 emission [kg]**
Concrete cores		9 970 275
Timber truss-system	Concrete cores	1 001 783
	CLT cores	470 229

From the table, it can be concluded that a timber truss system with CLT cores achieves the lowest CO_2. Note that, even though the calculations are based on rough estimates, a clear distinction can be made between the three variants. It is thus chosen to continue the design process with the timber truss system. The goal is to use CLT for the cores to achieve a minimal CO_2 emission. However, since it is uncertain whether this is structurally feasible, the CLT might be replaced with concrete cores. This will be further elaborated in Section 4.3.3.

The chosen stability system is shown in Figure 9. The principal is shown by the red and blue elements which are tension and compression elements respectively, when wind is coming from the side which is indicated by the arrows at the right.

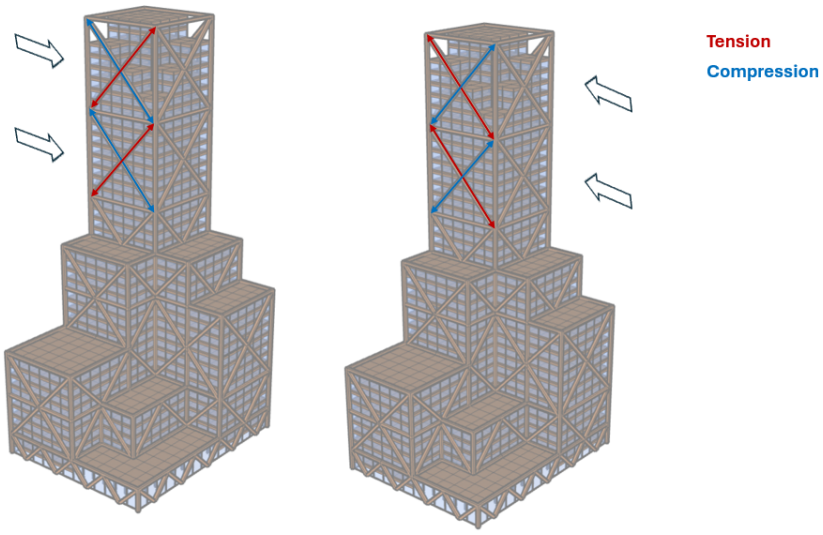


Figure 9: Timber truss stability principle

4 Structural design

4.1 Material choices

In line with the design goal to minimize the carbon footprint of the building, all material choices were evaluated primarily on their embodied CO_2 emissions. As established in Section 3, timber consistently outperforms conventional materials such as steel and concrete in terms of CO_2 performance. Due to its renewable nature and low embodied carbon, timber is particularly well-suited for reducing environmental impact in high-rise construction. Following the evaluation of different stability systems in Section **??**, it was concluded that concrete cores are not essential for ensuring lateral stability, as a timber truss system with diagonal bracing can adequately resist horizontal forces. Consequently, CLT (cross-laminated timber) cores will be used as the preferred solution to further reduce emissions, unless structural or risk limitations necessitate otherwise. The primary structural system for towers A to F consists of a timber column and beam frame. This approach ensures low embodied carbon throughout the main load-bearing system. The buildings mainhall, tower G, is constructed from concrete since the large beam spans and large column dimensions do not facilitate the use of timber. Concrete is only applied locally where necessary, such as for foundations or potential replacement of CLT cores if needed.

4.2 Loads and load cases

4.2.1 Load cases

In accordance with EN 1990 (Eurocode 0) and EN 1991 (Eurocode 1), the proposed high-rise office tower is classified as a Consequence Class 3 (CC3) structure. This classification applies to buildings where failure would result in significant risk to human life or lead to major economic or environmental consequences. As such, the structure must be designed with appropriate safety margins and robustness, and all load combinations must reflect this higher level of consequence. The load combinations used in this project are based on the partial safety factors and combination rules provided in EN 1990 and EN 1991. Both Ultimate Limit State (ULS) and Serviceability Limit State (SLS) combinations are considered. These combinations include permanent (G), live (Q), wind (W), snow (S) and accidental (A) loads and are appropriately factored. For ULS, both unfavourable and favourable conditions are included, and accidental load scenarios are also addressed. For SLS, characteristic, frequent, and quasi-permanent combinations are used to verify deformation and comfort requirements The applied load combinations are used to determine the dimensioning and structural integrity of the building in the following chapters. The full overview of applied load combinations is presented in Tables 3, 4, and 5.

Table 3: ULS load combinations – Unfavourable

Load case	**Combination**
Dead load	$1.35\,\mathrm{Gk} + 1.5\,\mathrm{Qk}$
Wind load	$1.35\,\mathrm{Gk} + 1.5\,\mathrm{Wk} + 1.1\,\mathrm{Qk}$
Snow load	$1.35\,\mathrm{Gk} + 1.5\,\mathrm{Sk} + 1.1\,\mathrm{Qk}$
Accidental loads	$1.35\,\mathrm{Gk} + 1.5\,\mathrm{Ak} + 1.1\,\mathrm{Qk}$

Table 4: ULS load combinations – Favourable

Load case	**Combination**
Dead load	0.9 Gk + 1.5 Qk
Wind load	0.9 Gk + 1.5 Wk + 1.1 Qk
Snow load	0.9 Gk + 1.5 Sk + 1.1 Qk
Accidental loads	0.9 Gk + 1.5 Ak + 1.1 Qk

Table 5: SLS load combinations

Load case	**Combination**
Characteristic	Gk + Qk
Frequent	Gk + 0.7 Qk
Quasi-permanent	Gk + 0.3 Qk
Wind	Gk + 0.5 Wk + 0.3 Qk
Snow	Gk + 0.2 Sk + 0.3 Qk

4.2.2 Loads

In accordance with EN 1991, all imposed actions relevant to the structural design of the high-rise building have been identified and are summarised in Table 6. Given the classification of the structure as a high-rise office and residential tower under Consequence Class 3 (CC3), appropriate load categories have been selected based on intended use and occupancy. Live loads for both office (Category B) and residential (Category A) areas are included, since the initial use of the building will be offices with the option to repurpose to residential, alongside climate-related actions such as snow and wind. The snow load of 0.7 kN/m^2 is adopted on the basis of the national annex to EN 1991-1-3. Wind actions are derived through calculation and yield average values of 2.1 kN/m^2 from the west and 1.7 kN/m^2 from the south, as detailed in Appendix A.4. Additionally, an accidental load of 1.0 kN/m^2 is applied in accordance with EN 1991-1-7 to ensure robustness under exceptional events.

Table 6: Overview of imposed loads

Load type	**Value [kN/m^2]**	**Reference / Note**
Live load (office, CC3)	3.0	EN 1991-1-1, Table 6.2 (Category B)
Live load (residential, CC3)	2.0	EN 1991-1-1, Table 6.2 (Category A)
Snow load	0.7	EN 1991-1-3, based on national annex
Wind load (west)	2.1	See Appendix A.4
Wind load (south)	1.7	See Appendix A.4
Accidental load	1.0	EN 1991-1-7, Section 4.3 – representative value

In addition to the variable actions, the dead load resulting from the floor system is considered as the permanent load (G). The floor consists of a multi-layered assembly of a CLT floor, an acoustic layer, a leveling layer, a finishing layer, and a suspended ceiling. The total calculated dead load amounts to 1,9 kN/m^2 as seen in Table 7.

Table 7: Dead load of the floor system

Floor build-up layer	**Thickness [m]**	**Density [kg/m^3]**	**Load [kN/m^2]**
Finishing layer	0.04	1800	0.71
Leveling layer	0.02	1200	0.24
Acoustic layer	0.03	150	0.04
CLT floor panel	0.16	500	0.79
Suspended ceiling	0.015	900	0.13
Total	0.265	–	**1.90**

The roof structure comprises several layers, including paving, drainage, insulation, and a CLT panel. Each contributes to the overall dead load applied as a permanent load (G) on the concrete beams. The total dead load from the roof build-up amounts to approximately 3.57 kN/m^2 as seen in Table 8.

Table 8: Dead load of the roof system

Roof build-up layer	**Thickness [m]**	**Density [kg/m^3]**	**Load [kN/m^2]**
Paving finish	0.06	2400	1.44
Drainage layer	0.05	1800	0.90
Protection layer	0.005	200	0.01
Waterproofing	0.005	1100	0.06
Thermal insulation	0.12	300	0.36
Vapour barrier	0.0005	900	0.005
CLT panel	0.14	500	0.79
Ceiling	0.015	900	0.13
Total	0.3955	–	**3.69**

4.3 Dimensioning

To begin the dimensioning of the structural elements, an appropriate grid size had to be established. Several grid configurations were evaluated, ranging from 5.0 × 5.0 m to 7.5 × 7.5 m for timber and 7.5 x 7.5 m to 15 x 15 m for concrete. The selection process was guided by the overarching design goal of minimizing embodied carbon, while also ensuring structural efficiency, spatial flexibility, and alignment with architectural intent. Ultimately, a 5.0 × 5.0 m grid was adopted for the timber structural system, as it offered the most favorable balance between material efficiency and modular repetition. For the concrete structure, the grid of 7.5 × 7.5 m was selected to reduce the dimensions of beams and, in turn, the required material.

Furthermore, for dimensioning the core and column, the building is separated into zones, as seen in Table 9.

Table 9: Division of the building into structural zones

Zone	Floors
1	1 to 4
2	4 to 10
3	11 to 17
4	18 to 24
5	25 to 32
6	33 to 38

4.3.1 Timber diagonals

The timber diagonals are dimensioned based on the calculations from the Excel sheet previously referenced in Appendix F. These initial dimensions will serve as input for the upcoming 3D structural analysis. Since the increased stability provided by the full 3D column arrangement is not accounted for in the 2D model, it is expected that the column dimensions can be reduced in the 3D configuration. This reduction will be determined through a process of trial and error, aiming to minimize the amount of timber used and thereby reduce the associated carbon emissions.

The same approach applies to the diagonal members and the floor elements. Not all intermediate floors are represented in the 2D calculations; additional floors exist between the diagonals in the actual design, and these will be incorporated into the 3D model. As a result, the dimensions of the diagonals are also likely to be reduced following further optimization in the 3D analysis.

Moreover, the 2D model assumes a single tower configuration. In contrast, the final design includes multiple towers, which increases the total facade area and, consequently, the moment of inertia. This additional stiffness will contribute to greater structural efficiency.

The input dimensions used in the 2D model are listed in Table 10 below after trial and error in the 3D model. These serve as preliminary values; the final dimensions will be refined and confirmed after completion of the 3D structural calculations.

Table 10: Diagonal dimensioning and embodied carbon values

Material	Dimensions [mm]	Volume [m^3]	Total Embodied Carbon [kg CO_2]
Timber (Glulam)			
GL32H	$1000 \times 1000 \times 22000$	22.00	22,352
GL28H	$1070 \times 1070 \times 22000$	25.08	25,487
GL24H	$1150 \times 1150 \times 22000$	29.26	29,722

4.3.2 Column and beam structure

Beams

The dimensioning of the beams was based on a comparative analysis of different glulam strength classes and concrete alternatives, with embodied carbon as the primary evaluation criterion. As shown in Table 11, multiple glulam classes were considered, but GL24H was ultimately selected due to its favourable balance between structural performance and low environmental impact. Although higher strength classes such as GL28H and GL32H offer increased capacity, their carbon footprint

is significantly higher, while the structural advantage in this application is marginal. The most carbon-efficient cross-section is chosen in the case of concrete as well, which is with class C35/C45.

The final timber beam dimensions were determined in alignment with common glulam manufacturing standards. Glulam beam widths are typically produced in increments of 40 mm to simplify fabrication and ensure compatibility with standard lamella sizes.

For the timber structure, GL24H glulam beams with dimensions of 320 × 560 mm were chosen based on modular sizing and low embodied carbon. In a concrete structure, beams of 270 × 1000 mm in concrete class C35/45 were selected to minimise both depth and environmental impact while maintaining structural efficiency. The in-depth calculations can be found in Appendix A.5.

Table 11: Beam dimensioning and embodied carbon values

Material	**Dimensions [mm]**	**Total Embodied Carbon [kg CO_2]**
Timber (Glulam)		
GL24H	320×560×5000	156.8
GL28H	320×560×5000	179.2
GL32H	320×520×5000	191.4
Concrete		
C25/30	270×1200×7500	668.2
C30/37	270×1100×7500	668.3
C35/45	270×1000×7500	658.1
C40/50	270×925×7500	674.3
C45/55	270×900×7500	710.8
C50/60	270×850×7500	705.7

Columns

The column design followed the same selection criteria as the beams, focusing on minimising embodied carbon while ensuring sufficient structural capacity. As presented in Table 12, three glulam strength classes were evaluated, with GL24H ultimately selected for its best environmental performance. The column dimensions of 720 × 720 mm are in line with standard widths of 40 mm.

For concrete columns, a range of strength classes from C25/30 to C50/60 was analysed again. As higher-strength concrete allows for smaller cross-sectional dimensions, the embodied carbon could be significantly reduced by switching to more efficient grades. Concrete class C50/60 was ultimately selected, as it enabled compact 270 × 270 mm columns while achieving the lowest CO emissions of all options assessed. The concrete column dimensions align with the beam dimensions to facilitate smoother connections.

Therefore, for timber GL24H is chosen across all zones and the dimensions are found in Table 13. The concrete columns are C50/60 of 270 x 270 mm. The in-depth calculations can be found in Appendix A.6.

Table 12: Column dimensioning and embodied carbon values

Material	**Dimensions [mm]**	**Total Embodied Carbon [kg CO_2]**
Timber (Glulam)		
GL24H	720×720×4000	362.9
GL28H	720×720×4000	414.7
GL32H	680×680×4000	425.4
Concrete		
C25/30	400×400×8000	352.0
C30/37	360×360×8000	311.0
C35/45	320×320×8000	266.2
C40/50	300×300×8000	259.2
C45/55	285×285×8000	253.4
C50/60	270×270×8000	239.1

The timber columns, as seen above, are for zone 1, as seen in Table 9. The dimensioning of the columns increases downwards due to the increasing axial load. In Table **??** the dimensioning for all zones is seen.

Table 13: Column dimensions by structural zone

Zone	**Floors**	**Columns [mm]**
1	1–4	720 × 720
2	4–10	720 × 720
3	11–17	720 × 720
4	18–24	640 × 640
5	25–32	600 × 600
6	33–38	520 × 520

4.3.3 CLT cores

To minimize the environmental impact of the structural cores, only CLT was considered, as concrete is not necessary to ensure global stability, as discussed in Section 3.3. Several timber grades and glulam alternatives were evaluated based on strength and embodied carbon. As shown in Table 14, standard C24 CLT panels exhibited the lowest carbon footprint while providing sufficient stiffness and strength for the core walls. Therefore, C24 was selected as the material for all CLT cores throughout the building. The in-depth calculations are presented in Appendix A.7.

Table 14: Comparison of materials for CLT cores

Material	**Thickness [mm]**	**Total Embodied Carbon [kg CO_2e/m]**
C24	220	31.9
C30	220	35.2
C35	220	39.6
GL24H	220	38.8
GL28H	220	44.0
GL32H	200	46.0

The CLT core thickness varies by zone to reflect the decreasing loads toward the top of the building. As shown in Table 15, panel thickness is reduced from 220 mm at the base to 180 mm at the top, optimizing material use without compromising structural performance. The ply buildup of the CLT core elements is shown in Table 15 and is based on manufacturing standards.

Table 15: CLT core thicknesses and ply build-up by structural zone

Zone	Floors	Core thickness [mm]	Ply build-up
1	1–4	220	7-ply (40 + 28 × 5 + 40)
2	4–10	220	7-ply (40 + 28 × 5 + 40)
3	11–17	200	7-ply (40 + 24 × 5 + 40)
4	18–24	200	7-ply (40 + 24 × 5 + 40)
5	25–32	200	7-ply (40 + 20 × 5 + 40)
6	33–38	180	7-ply (40 + 20 × 5 + 40)

4.4 Structural calculations

In the following section the dimensioning of the structural elements is checked based on unity checks.

4.4.1 Columns

The unity checks for the timber columns (Table 16) show that both compressive and buckling capacities are well within safe limits across all zones. The highest unity check for compression is 0.434, while the highest for buckling is 0.480, indicating a conservative yet efficient structural design. These results confirm that the selected cross-sections offer sufficient strength while maintaining a low material impact, aligning with the project's sustainability goals. The reason of the results being conservative is the fire safety, which will be mentioned in a later stage.
Similarly, the concrete columns (Table 17) show unity check values far below the limit, with the highest buckling check reaching only 0.579. There are three separate columns which each of which has different loading from the roof area and the number of beams connected therefore, the loads vary. This demonstrates that the dimensioning of the concrete sections—despite their limited use in the overall structure—is more than adequate. Overall, the low unity check values in both timber and concrete columns reflect a method to conservative for dimensioning, leaving room for optimization while ensuring safety and compliance with design goals. The in-depth calculations can be found in Appendix A.8.

Table 16: Unity checks for timber columns per zone

Zone	Compressive UC	Buckling UC
1	0.434	0.480
2	0.388	0.475
3	0.308	0.377
4	0.289	0.250
5	0.214	0.302
6	0.131	0.220

Table 17: Unity checks for concrete columns

Column	Compressive UC	Buckling UC
1	0.309	0.579
2	0.232	0.399
3	0.155	0.220

4.4.2 Beams

The structural performance of both timber and concrete beams was evaluated in terms of bending and deflection. As shown in Table 18, the GL24H timber beams reach a bending unity check of 0.463 and a deflection check of 0.430, both well below the limit. This indicates that the selected timber section dimensions are sufficient for load-bearing while remaining material-efficient. The timber is again lower due to fire safety considerations.

Concrete beams (C35/45) show even lower unity checks, with values of 0.163 for bending and 0.662 for deflection. Despite the higher stiffness of concrete, the deflection ratio is slightly closer to the limit, though still acceptable. Overall, the beam design for both materials meets structural requirements with a margin for refinement or optimization. The in-depth calculations are presented in Appendix A.8.

Table 18: Unity checks for timber and concrete beams

Material	Bending UC	Deflection UC
Timber [GL24H]	0.463	0.430
Concrete [C35/45]	0.163	0.662

4.4.3 Cores

The unity checks for the axial perfomance of the cores are well below the 1.0 for the unity checks, as shown in Table 19, this is due to fire safety requirements, see Section 5.3.2, for high-rise timber structures, the CLT core walls must meet a minimum wall thickness to ensure sufficient charring resistance over time. This results in relatively thick wall sections even in higher zones where structural loads are significantly lower. The low UC values confirm that the dimensioning of the CLT walls is governed primarily by fire resistance rather than structural demand, leading to a conservative design. The in-depth calculations can be found in Appendix A.8.

Table 19: CLT core wall compression unity checks by zone

Zone	Wall Thickness [mm]	Compressive UC
1	220	0.194
2	220	0.173
3	200	0.151
4	200	0.112
5	200	0.073
6	180	0.037

4.4.4 SCIA Engineer calculations

To verify the interaction and performance of all structural elements within a unified model, two SCIA Engineer models were developed. The first model is a simplified version focusing solely on the exoskeleton system to assess its contribution to the overall stability. The second model includes all key structural elements integrated into a single tower for a more comprehensive analysis. Due to performance limitations in SCIA, not all seven towers of the full building were modeled, as this significantly slowed down the software.

The SCIA Engineer results are presented in Appendix A.9. The first model includes all primary structural elements; beams, columns, floors, and diagonals, and was used to assess the overall structural performance of the tower. This comprehensive model allows for the evaluation of global stability, column compression forces, and floor load-bearing capacity under applied loads.

As shown in the results, the addition of adjacent towers significantly reduces overall deflection—cutting it by half compared to the single-tower model. Initially, 1.5 $\times$ 1.5 meter diagonals were used, but through a process of trial and error, it was found that 1.0 $\times$ 1.0 meter diagonals provided sufficient stability, allowing for material savings.

The simplified model, which focuses primarily on the exoskeleton's role in stability, showed similar deflection behavior. However, a new challenge was identified: differential deflection between adjacent towers. Without proper connection at the roof level, these differences could lead to cracking or separation in the roof structure. Figure 10 illustrates this issue on the right, along with the proposed solution on the left. To resolve this, additional roof-level diagonals must be introduced to ensure that forces from the taller tower can be effectively transferred to the adjacent structure.

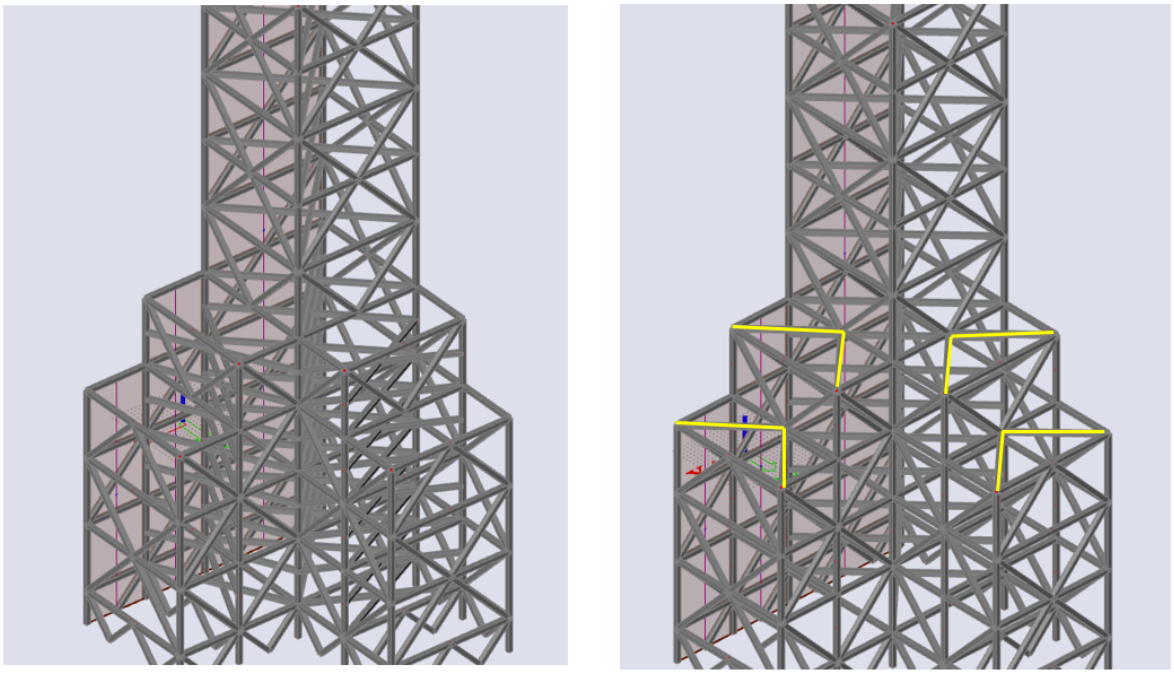


Figure 10: Connection detail beam-column-diagonal

4.5 Structural connections

In Figure 11, a relevant connection detail of the timber exoskeleton of the building is shown below. The connection is between the beam, column,n and diagonal of the exoskeleton. This connection method draws inspiration from the renowned structural solutions applied in the Mjøstårnet building in Norway, which is currently one of the tallest timber buildings in the world.

The detail illustrates a robust and highly engineered timber node where multiple glulam members intersect, joined together with steel plate connectors and numerous dowel or bolt fasteners. This approach ensures effective load transfer between intersecting beams and columns, providing both strength and stiffness necessary for high-rise timber structures.

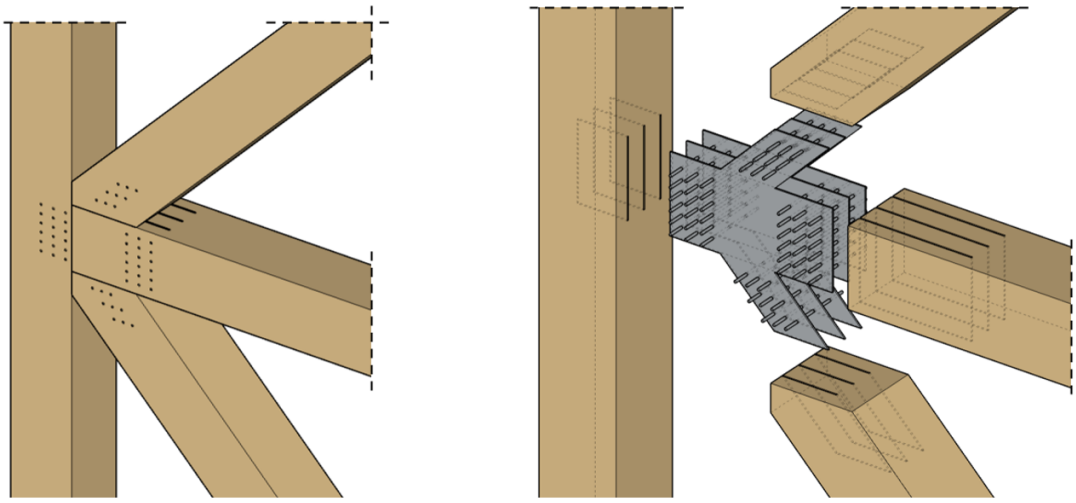


Figure 11: Connection detail beam-column-diagonal

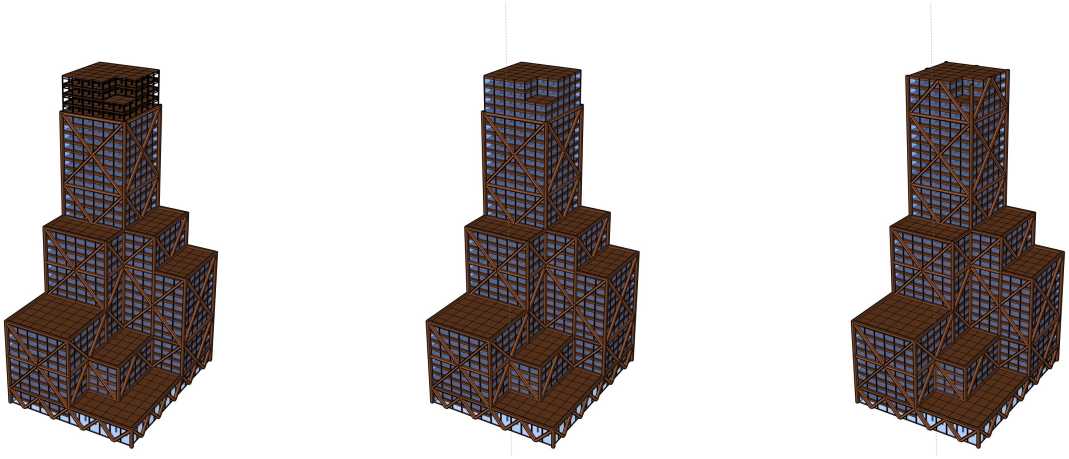
Similar connection principles can be seen in Mjøstårnet's exoskeleton, where steel plates and embedded connectors are used to join timber elements while accommodating complex load paths, including compression, tension, and shear forces (Figure 12). The use of steel connectors within timber nodes helps to reduce issues related to timber's anisotropic properties and enhances the overall ductility and resilience of the structure.

The detail based on the Mjøstårnet design uses proven technology to achieve a reliable and high-performing timber exoskeleton capable of resisting the significant demands of a tall building in the same range as the design of the building is this project. This includes lateral wind loads and vertical gravity loads and maintains architectural aesthetics as well.

Figure 12: Connection detail beam-column-diagonal Mjøstårnet (Sweco, Mjøstårnet)

4.6 Construction plan

To realize the building, a construction plan is set up, in which the construction order is elaborated. This plan is visualized in Figure 13 for stories 31-14. The full plan can be found in Appendix A.10. From the Figure, it can be found that the construction is built up in parts of four stories, as each diagonal spans over four stories. Repeated for all sections of four, first the column and beam structure will be placed, followed by the facade, and finally the diagonals are placed. Even though this method will allow for the least amount of instability in comparison to alternatives like building all towers individually, a temporary structure is needed over the four floors before the diagonals are placed. Since the diagonals are placed per four floors, the temporary structure can be re-used over the entire building process.



(a) (b) (c)

Figure 13: Construction plan of stories 31-34, showing (a) the construction of the column-beam structure, (b) the addition of the facade, and (c) the construction of the timber diagonals.

5 Final design

5.1 Architectural design

The architectural design of the building is in unison with its urban environment within the VDMA area of Eindhoven, aligning with the city's evolving skyline through stepped design. This stepped configuration not only enhances the building's visual integration with its surroundings but also contributes to reduced wind loads and improved pedestrian comfort by mitigating downdraft effects at street level, which is aided by the external diagonals breaking downward wind draft. A render of the tower is shown in Figure 14, and an integrated view in the city is seen in Figure 15. As seen in Figure 15, it stands out in the current environment but should blend in with the current plans of Eindhoven, for example, stationsgebied knoop XL. A physical model was handcrafted to give a physical impression of the design; these can be found in Appendix A.11.

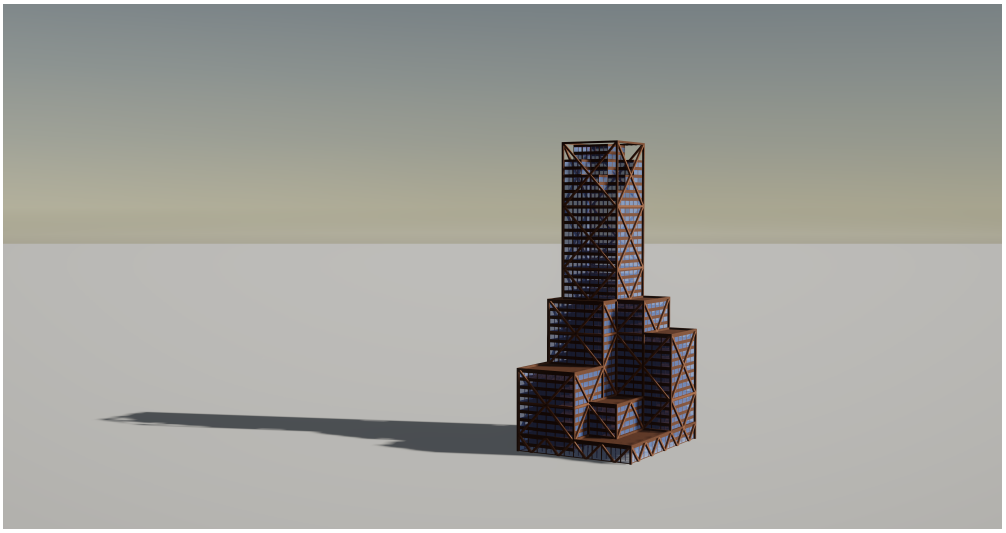


Figure 14: Rendered imagine final architectural design

Figure 15: Rendered imagine final architectural design in the context of Eindhoven

The entrance hall forms a prominent and inviting base to the complex. It is designed with a concrete column and beam structure, utilizing a 7.5 × 7.5 meter grid, and features a generous free height of 8 meters. This creates a spacious and open atmosphere at the ground level. Positioned directly above the entrance is a terrace that introduces a soft spatial transition between the robust plinth and the lighter timber tower structure.

The regular towers are constructed using a timber column and beam system, with integrated CLT cores that serve both structural and fire safety functions. Each floor provides a free height of 3.2 meters, including the floor and beam system, ensuring flexible and comfortable interior spaces. Stability is further enhanced by timber diagonals placed within the façade. These not only provide lateral resistance but also help in managing wind behavior, reducing the downdraft experienced by pedestrians around the base of the building. Tower F, as mentioned in Table 1, also has a roof terrace at a height of 24 meters, which is mostly out of the wind due to the tower's configuration. The stepped design is also implemented in the highest tower, where the top floors have a stepped design in a similar configuration to the rest of the design.

5.1.1 Open floorplans and flexibility

The building is designed for office spaces. To align with current trends, it is decided to keep the floor plan as open as possible. Figure 16 shows an example of a floorplan located on floor 10. An enlarged view and the complete set of floorplans can be seen in Appendix A.12. As can be seen in the figure, the office floors are completely open, with the exception of areas reserved for facilities such as sanitary rooms and technical rooms. These rooms will be placed towards the inside of the building to ensure daylight in the spaces where it is needed most.

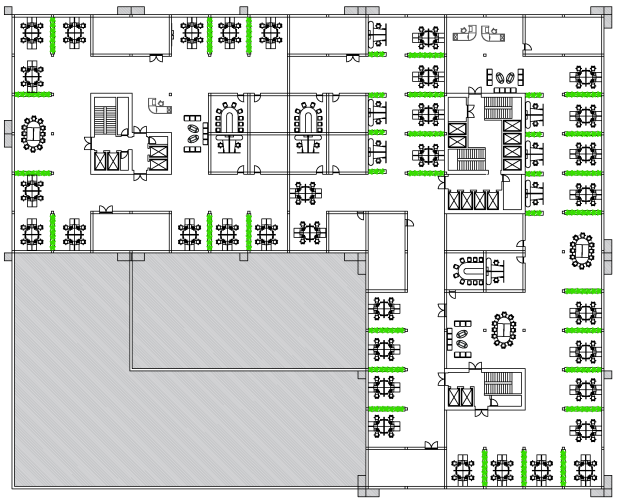


Figure 16: Exemplary floorplan floor 10

Not only does this openness create a comfortable work environment in terms of lighting and connection to coworkers, but it also allows for flexibility in the building's function. The column and beam structure allows for the free placement of timber frame walls, which can be used to create compartmentalization, for example, for apartments. The depths of these potential spaces are limited to a maximum of 10 meters due to the placement of the cores, ensuring daylight is accessible

everywhere in the building. When changing the function from offices to apartments, the density of users decreases, meaning the transport systems remain sufficient.

5.1.2 Daylight

The column and beam structure yields an open workplace, allowing daylight to enter freely. The building's design ensures access to daylight within two grid sizes, adding up to 10 meters in total. Having this daylight entrance was also considered a requirement for the potential repurposing of the building into apartments, as daylight requirements for living spaces are strictly set.

To ensure a comfortable lighting situation and stimulate the non-image-forming effects caused by sufficient daylight while working, a daylight simulation is performed. This simulation assumes a worst-case scenario in which a room is one grid size wide and two grid sizes deep (5 x 10 m), meaning no light enters from the side; only from the window. Due to fire safety reasons, as elaborated in Section 5.3.2, not the entire facade is considered glass; however, windows are placed with a vertical space of 0.9 m. This results in a total glass area of 11.3 m^2. The results of the simulation can be seen in Figure 17.

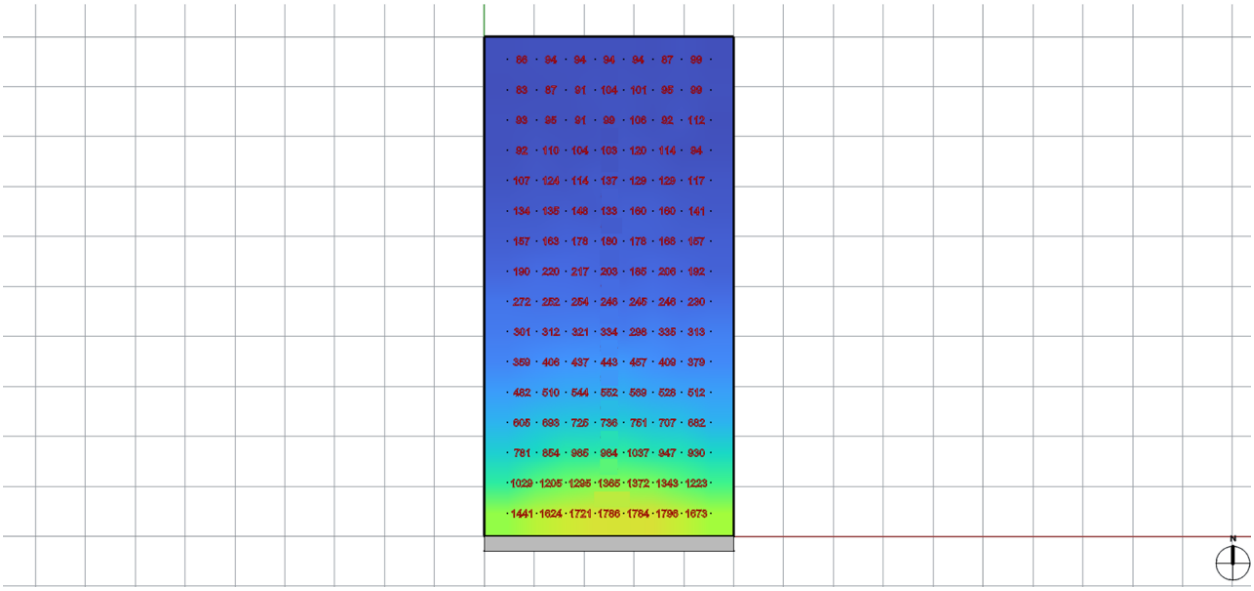


Figure 17: Results daylight simulation

As can be seen, sufficient daylight enters the space throughout. For an office space, the requirements are set to achieve 500 lux on a working surface. To fulfill this requirement, it is recommended to place desks close to the window. The space towards the back can be used for storage or, in the case of an open office space, for a passageway. All spaces that do not have direct access to a window will not be assigned office functions but rather non-daylight-requiring functions, such as sanitary and technical spaces. In the event that the building is converted into apartments, it is recommended to place non-living spaces, such as bathrooms, towards the interior of the space. The annual sunlight area is 35%, vastly exceeding the requirements of 10% for living spaces and 2.5% for office spaces.

5.2 Transport system

To ensure safe and efficient evacuation and daily usability, a well-organized vertical transport system has been designed. The layout includes **three vertical cores**: A, C1, and E1, where A has three sub cores A1, A2, A3. These cores are distributed across the towers to serve different zones and heights, as visualized in Figure 18, where the left image shows a south-to-north view and the right image shows a west-to-east view.

The cores operate across the following levels:

- **Core A1** (red): full-height, serving only 19-38.
- **Core A2** (green): serves *floors 1–18*.
- **Core A3** (yellow): serves *floors 1–6*.
- **Core C1** (blue): serves *floors 1–14*.
- **Core E1** (purple): serves *floors 1–10*.

This system is based on an occupancy rate of **7 persons/m^2**, in accordance with NEN-EN 1991-1-1, Table B.2, which provides representative crowd densities and live loads for office buildings. The overlapping vertical ranges ensure redundancy and efficiency in elevator usage while minimizing wait times and optimizing shaft allocation. It allows everyone to be transported throughout the building in accordance with NEN-EN 1991-1-1.

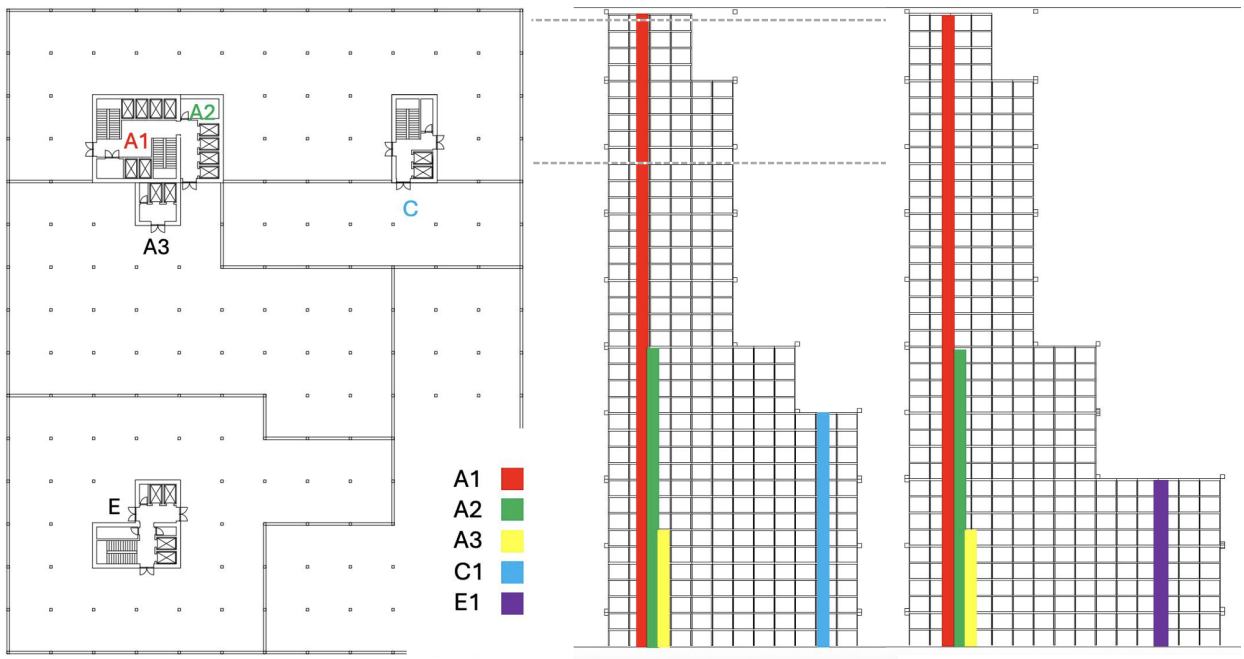


Figure 18: Overview of the vertical transport cores and their distribution per tower

5.3 Risk analysis

5.3.1 Acoustics

A significant risk in using timber floor and wall systems is the acoustics, as timber easily transfers vibrations. To prevent both airborne sound and impact sound transfer, remedial measures are incorporated into the design. These solutions are shown in Figure 19. An enlarged version is shown in Appendix A.13. Note that this detail presents a general solution for a wall-floor connection and will thus not be applied everywhere.

To prevent the impact sound transfer through the walls, a double wall structure is installed, separated by a cavity. This cavity will directly stop vibrations from passing through the timber structure. To prevent airborne sound transfer, the walls are insulated using a material with a high absorption coefficient, such as mineral wool. As the mineral wool has a larger area in comparison to the timber framework, it is assumed that this measure provides a sufficient reduction in airborne sound transfer through the walls.

To prevent airborne sound transfer through the floor systems, suspended ceilings with mineral wool will be applied. The vertical impact sound transfer in the floor systems will be prevented using floating screed flooring. As shown in Figure 19, dampening material is placed on both the bottom and the edge of the flooring, which prevents vibrations from traveling into the timber structure. In addition to the separation of the flooring, mineral wool is applied between the connection of flooring elements to prevent horizontal traveling of vibrations through the floor elements. Note that some floors transfer horizontal forces between the diagonals in the exoskeleton. In these floor elements, this solution will not be applied. Local solutions will be found for the spaces surrounding these floor elements.

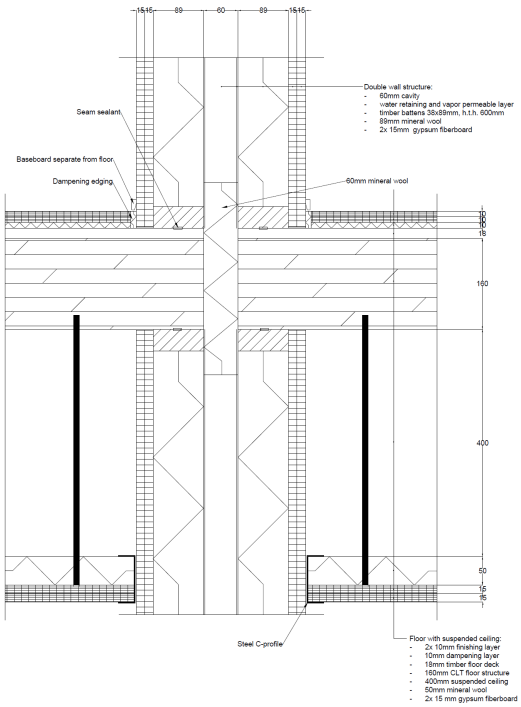


Figure 19: Detail acoustic remedial measures

5.3.2 Fire safety

Fire safety is a critical aspect in the design of high-rise timber buildings, particularly due to the combustibility of the material. In accordance with the requirements for consequence class 3 (CC3), strict measures are implemented to ensure adequate safety in the event of a fire. The fire strategy encompasses both active and passive measures, with compartmentalization playing a crucial role. Fire compartments are designed to limit the spread of fire and smoke, allowing occupants to evacuate safely and aiding the intervention of emergency services. According to **NEN 6060**, the maximum compartment size in timber buildings is limited to **1000,m²**, as illustrated in the compartmentalization plan in Figure 20, in order to ensure sufficient containment and manageability in case of fire.

Figure 20: Fire compartmentalization strategy showing zoning per floor to comply with the 1000 m^2 maximum area limit per compartment, as required by NEN 6060.

In addition to compartmentalization, the structural elements must retain their load-bearing capacity during exposure to fire. For buildings in CC3, a fire-resistance rating of **R120** applies, meaning the structure must withstand fire for 120 minutes. According to *NEN-EN 1995-1-2*, this is evalu-

ated using effective charring rates and a zero-strength layer that accounts for material degradation near the char line.

For the CLT cores, which are expected to face fire exposure from only one side, an effective charring rate of 0.65,mm/min is assumed. Including a 7,mm zero-strength layer, this results in a sacrificial depth of 85,mm. Glulam columns and beams are considered exposed on two sides and use a charring rate of 0.70,mm/min, resulting in a sacrificial layer of 91,mm. The remaining cross-sectional area is then checked to ensure structural stability under fire conditions. This is done by using the serviceability limit state (SLS). In Appendix A.14, a detailed calculation about how the charring depth influences the material dimensioning is found.

It should also be noted that it was chosen not to make the full facade out of glass but to add fire-resistant facade elements to provide vertical distance between the windows. No exact calculations were performed; however, based on common practice, a distance of 0.9 m was chosen. It is expected that this will reduce the chance of vertical fire spread over different fire compartments, making fire-resistant glazing unnecessary. This decision aligns with the chosen design goal of reducing CO_2 emissions in material use.

Finally, it should be noted that it is standard practice to use concrete cores in medium to high-rise timber buildings; however, according to NEN 6060, the current design is allowed.

5.3.3 Maintenance

To reduce degradation and thus maintenance, the exoskeleton is designed to prevent water accumulation and long-term water retention. Excessive water accumulation can be prevented with a good drainage system. Water needs to be drained from the horizontal beams, as the risk of accumulation is the largest in these elements. The water also needs to be drained from the connection points of diagonals. To prevent long-term water retention, the exoskeleton cannot be placed directly against the facade; instead, an air gap is required to allow ventilation around all beams. This ventilation can allow for a faster drying time of the trusses, which is needed to lower the moisture content in the timber to under 20%, the threshold for molding. However, considering the wet environment in which the building will be placed, it is expected that this threshold of molding will be surpassed for a considerable amount of time.

Therefore, even though these measures provide some improvement in terms of reduced maintenance, they are not expected to suffice. Due to expected UV and moisture degradation, repeated treatment of the timber exoskeleton is needed. As a structure of these dimensions as an exoskeleton has not yet been applied in practice, it is difficult to assess the needs in terms of maintenance precisely, but it is expected that, even with repeated treatment of the timber and the changes in design for water drainage and ventilation, it will not be sufficient to ensure a lifespan of more than 50 years. Even though entirely replacing the timber diagonals will still result in less CO_2 emission than an equivalent steel system (see Appendix A.15), this is not a feasible option for the design).

5.4 Conclusion in regards to the design goal

The total CO_2 emission of the building materials is 122.50 $kgCO_2/m^2$. Although it is challenging to find an exact reference source due to the inclusion of only material use in this calculation and the building's magnitude, a few reference sources are found to determine whether the goal can be con-

sidered achieved. According to a report by Bloomberg Philanthropies [3], apartment buildings and multi-story office buildings are estimated to emit a minimum of 490 kg-CO_2/m^2 for both material production and construction-related emissions. Similarly, a study by Luo et al. [4] estimates the median CO^2 emission for the material production and construction of an office building in China to be 533.67 kg/m^2. Even when considering the inclusion of construction-related emissions in these studies, it can still be concluded that the designed timber office building has a relatively low CO_2 emission from material use.

It should also be noted that, regardless of the comparison to literature, all impactful decisions were made in accordance with the design goal. This resulted in conventional solutions being set aside to achieve lower CO_2 emissions, which in itself can be considered a successful result for the design goal.

6 Discussion

6.1 Feasibility of the design

The goal of this assignment was to minimize the CO_2 emission from material use in the building. Choosing this design goal has led to setting aside conventional solutions and taking risks in the structural design. It has prompted thinking outside the box and critically evaluating the application of design and calculation rules to this specific design. Although the challenges were welcomed, as mentioned in Section 5.4, the expected durability issues with the timber diagonals render the design unfeasible. To continue developing the presented design plan, alternatives should be explored that minimize the main structural elements' susceptibility to deterioration.

6.2 Proposed alternatives

Even though the timber exoskeleton was chosen as a final design, two alternatives are proposed to prevent the arising durability issues by protecting the external timber. In further research or design, these alternatives can serve as a starting point to reevaluate the most appropriate design choices for the tower.

- The first alternative is implementing a double skin facade, which consists of a two-layer facade with a cavity in between. Using this system, the structural design could remain similar, as a new layer could be added to the external side of the diagonals. However, careful consideration needs to be put into the ventilation between the two facade layers, as the diagonals block direct ventilation. If implemented correctly, the cavity could act as an insulating buffer zone, reducing heat gain and loss, thus improving energy efficiency and reducing CO_2 emissions. However, the additional material would increase CO_2 emissions solely from material use, which contradicts the goal of this assignment. Another issue is presented by the wind flow downwards from the side of the building. Currently, the diagonals disrupt any downward wind stream, compromising pedestrian comfort. In further design, remedial measures should be implemented to prevent this discomfort.
- The second alternative is moving the now exoskeleton behind the facade, which will also result in environmental protection of the timber diagonals. Similarly to the double skin facade, this will result in a smooth surface over the facade, meaning remedial measures should be implemented to prevent pedestrian discomfort. Additionally, the structural model of the exoskeleton will need to be reevaluated, as it will now interact with the column-beam structure

in the tower and the floor systems. Additionally, the detailing will become more complex, as the diagonals connect to the interior structure more often now. From the performed calculations elaborated in Section 3.3, it is found that this design would suffice structurally. To conclude, assuming that further research will keep in mind the set design goal of minimizing CO_2 emissions from material use, this alternative is believed to have the most potential for further research.

7 Personal reflections

7.1 Reflection Vigo de Groot

Reflecting on this project, I found it to be a challenging but rewarding experience that offered valuable insights into both structural design and the broader implications of working with sustainable materials in high rise buildings. While our ambition to use timber as the primary structural material introduced limitations, it also pushed us to think more critically about feasibility, detailing, and design integration.

One of the most useful aspects of our process was the distinction we made between 2D and 3D structural analysis. Early in the project, using 2D calculations allowed us to make quick decisions about sizing and stability principles. This was extremely helpful for exploring different design options without getting bogged down in complex modelling too early. It gave us the flexibility to compare alternatives and understand the influence of different geometries, materials, and connection strategies on performance.

Once we had a more refined concept, the 3D SCIA model allowed us to evaluate the behaviour of the structure as a whole. It confirmed many of our assumptions from the 2D analysis, but also highlighted where adjustments were needed, particularly in terms of deflection between towers and the need for continuity in the roof structure. Although a full 3D model from the start would have been ideal in theory, in practice it would have slowed our workflow significantly. I've learned that a phased approach, starting with 2D and moving to 3D strategically, is often the most efficient way to manage time while ensuring accuracy.

A key learning moment for me came during the design of the connections. Initially, I underestimated their complexity, assuming they could be solved once the main geometry was fixed. However, I soon realised that connection design is a central part of structural behaviour, especially in timber, where force transfer, fire protection, tolerances, and prefabrication all play a role. This taught me that connections shouldn't be an afterthought, they need to be considered from an early stage to avoid major rework later on.

Another important insight came from analysing the interaction between towers. The addition of roof-level diagonals between adjacent towers turned out to be essential to avoid differential deflections that could otherwise cause structural and façade issues. Without these connecting elements, the individual towers would behave too independently, leading to stress concentrations and potential cracks in the roof. This detail became a crucial component in achieving a coherent overall structural behaviour.

From a group perspective, I believe our collaboration was effective despite some logistical challenges. I appreciated the open discussions we had, particularly when aligning our personal preferences with

the project's sustainability goals. The fact that we continuously questioned our assumptions and revisited earlier decisions, even if it cost us some time, ultimately led to a more cohesive and thoughtful design.

In the end, although the feasibility of our final structure remains limited, the experience taught me to work at the intersection of technical design, sustainability, and architectural intent.

7.2 Reflection Manon Hanssen

In general, I believe our process throughout the project was effective. As a group, we had a rocky start with the scheduling of the course and tutor sessions, as two of our group members were unavailable during the chosen time slot. However, we were able to make it work by having one student attend the sessions and holding regular meetings on an alternative day. This way, we were able to present our interim design at the beginning of the project and get additional feedback during the midterm presentations. The project's timeline made it difficult for us to stay entirely focused around the time of the exams in quarter 3. However, at the start of quarter 4, we regrouped and continued with the project.

Our design goal has had a significant impact on our process. Throughout the project, we realized that our design goal significantly limited our options. What started as a motivated decision to contribute to a more sustainable built environment resulted in a severe limitation in the design choices. When investigating stability principles, we noticed that all three of us were leaning more towards using a concrete core. However, from a CO_2 perspective, we could not justify this over using a timber exoskeleton. Even though it was obvious from our design goal which decision we had to make, our indecisiveness cost us a lot of time midway through the project. We got stuck on deciding between our preference and the design goal, so we tried validating a design with a concrete core from a CO_2 perspective. Ultimately, we realized that our only viable option was the timber structure. At this point, we shifted our mindset and fully embraced the design goal.

It should also be mentioned that, having chosen an unconventional system, we were pushed to think outside the box and the current regulations. We were able to explore why the more sustainable material choices we were making were not common practice and found many puzzles that needed solving to arrive at a final design. Although we may have been hesitant to make definitive decisions at times during the project, I enjoyed the in-depth discussions about the precise working and application of our structural system. I also really enjoyed how the design goal ultimately impacted our design on a much more integrative level than I initially expected. I thought the CO_2 reduction would only result in using timber instead of concrete and steel. However, we ultimately implemented the design goal in many different decisions, such as grid size and dimensioning. We even found that the higher CO_2 emission of fire-resistant glazing compared to normal glazing led to an increase in the vertical distance between the windows, which in turn impacted our daylight calculations.

Looking back at the project, considering that it resulted in an infeasible design, there are definitely things I would have done differently. As mentioned earlier, we spent a considerable amount of time deliberating over our primary stability principle. If we had made that decision sooner, we could have identified the shortcomings of our design more quickly and thus found solutions and implemented them more effectively. I don't regret the design goal we chose, but we did let it impact our process too much. Looking back, we should have either thought about the goal in more detail with

design ideas in mind, or not had a specific design idea in mind that did not align with the goal. In the end, I'm really glad we ended up going for a design that was in line with the goal, even though we did not manage to make it feasible.

As already discussed in Section 6.2, we considered some solutions to address the shortcomings of the design. I personally believe that, in further developing the project, moving the timber trusses to the inside would be the most feasible option. I think the mistake we made was changing out a conventional material like steel for timber to reduce CO_2 without considering its inherent properties, and that is definitely a mistake that I have learned from.

7.3 Reflection Jeroen Simons

Reflecting on the project, I consider the process to have been both demanding and educational. The initial phase presented some challenges in terms of group coordination, particularly due to scheduling conflicts, but we eventually established a productive routine with weekly meetings. After the midterm, progress slowed somewhat due to academic workload during the exam period. However, we were able to regain momentum in Quarter 4 and continue with a clear focus. While our meeting structure generally kept the group aligned, I recognise that a more proactive attitude—both individually and collectively—could have improved our workflow during key moments.

The design goal of reducing embodied carbon strongly influenced the entire process. While it gave structure to our decisions, it also introduced significant constraints that limited design freedom, particularly regarding material and stability choices. This tension was at times difficult to navigate, but it provided valuable insight into the complexity of sustainable structural design. I learned that sustainability goals often extend beyond material selection and influence other technical aspects, such as fire safety requirements, spatial layout, and even façade design.

Choosing a fully timber structure appeared straightforward in theory, but proved to be complex in practice. The implications of fire resistance, structural limitations, and regulatory constraints became increasingly apparent as the project developed. In particular, the fire safety requirements imposed by height thresholds highlighted how relatively minor regulatory boundaries can significantly impact overall feasibility. These constraints forced us to explore alternative solutions and reconsider initial assumptions.

The final phase of the project also brought logistical challenges, particularly in the physical model construction. A last-minute issue with the laser cutter delayed the process, resulting in a rushed fabrication phase. In hindsight, allocating more time for model production and anticipating such setbacks would have been prudent. Additionally, I came to understand the importance of maintaining well-structured and adaptable Excel sheets from the outset. As our calculations evolved, a clearer setup would have made iteration and verification far more efficient.

Although there are several aspects I would approach differently in retrospect—particularly in terms of early design decisions and project planning—I found the project to be an important learning experience. It sharpened my understanding of the interplay between sustainability goals, structural logic, and practical constraints. Working in detail with timber was a new and rewarding experience, and the challenges we encountered have significantly informed my perspective for future design work.

Finally where the design could have been improved but time was lacking was the importance of checking shear in timber elements, currently this isn't checked which could lead to major flaws in the design, due to the lack of time none of us was able to check this anymore. Also a lack of understanding within timber design to not having followed the course timber had lead to the late realization of the importance of shear in timber construction.

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[4] Zhixing Luo, Yujie Cang, Nan Zhang, Liu Yang, and Jiaping Liu. A quantitative process-based inventory study on material embodied carbon emissions of residential, office, and commercial buildings in china. *Journal of Thermal Science*, 28:1236–1251, 2019.

A Appendices

A.1 Core calculation

Core calculation

Concrete class	Elastic modulus	Unit
C50/60	37000	N/mm^2

Dimensions	Width	Height	Unit
Outer dimensions	12	12	m
Inner dimensions	9	9	m
Thickness core	1,5	1,5	m
Length core (L)	150	m	

Properties		Unit
Moment of inertia (I)	1181,25	m^4
Weighted moment of inertia (EI)	13111875000	kNm^2
Windload (q)	54	kN/m

Results	Amount	Unit		
Displacement (w)	0,260617761	m		
Maximum displacement	0,3			
Maximum moment	607500	kNm		
Kurvature	4,6332E-05	m^{-1}		
Strain (outside of concrete)	0,000277992	-		
Strain (outside of concrete)	0,000277992			
Compression (outside of concrete)	10285,71429	kN/m^2	10,2857	N/mm^2
Tension (outside of concrete)	10285,71429	kN/m^2	10,2857	N/mm^2

Concrete class	Elastic modulus	Unit
C50/60	37000	N/mm^2

Dimensions	Width	Height	Unit
Outer dimensions	15	15	m
Inner dimensions	13,8	13,8	m
Thickness core	0,6	0,6	m
Length core (L)	150	m	

Properties		Unit
Moment of inertia (I)	1196,4672	m^4
Weighted moment of inertia (EI)	13280785920	kNm^2
Windload (q)	54	kN/m

Results	Amount	Unit		
Displacement (w)	0,257303108	m		
Maximum displacement	0,3			
Maximum moment	607500	kNm		
Kurvature	4,57428E-05	m^{-1}		
Strain (outside of concrete)	0,000343071	-		
Strain (outside of concrete)	0,000343071			
Compression (outside of concrete)	12693,62002	kN/m^2	12,69362002	N/mm^2
Tension (outside of concrete)	12693,62002	kN/m^2	12,69362002	N/mm^2

A.2 Truss calculation

Truss calculation

Dimensions		
Cross sectional height column	1,5	m
Cross sectional width column	1,5	m
Length core (L)	150	m
width building	30	m

Properties		Unit
Weighted moment of inertia (EI)	13365000000	kNm^2
Windload (q)	54	kN/m
Elastic modulus timber GL32h	13200	N/mm^2

Bending		
Displacement (w)	0,255681818	m
Maximum displacement	0,3	m

Dimensions		
Cross sectional height floor	1,75	m
Cross sectional width floor	1,75	m
Cross sectional height diagonal	1,75	m
Cross sectional width diagonal	1,75	m
Height of one story	3,6	m
Length diagonal	30,21522795	

Properties		Unit
GA	2115241,507	kNm^2

Racking shear		
Displacement (w)	0,287201248	m
Maximum displacement	0,3	m

A.3 Comparison CO_2 emission of different stability principles

Concrete cores

Main core			*Core 2*			*Core 3*			*Core 4*		
Height tower	150	m	Height tower	75	m	Height tower	50	m	Height tower	30	m
Dimensions			Dimensions			Dimensions			Dimensions		
Width	12	m	*Width*	8	m	*Width*	10	m	*Width*	6	m
Depth	12	m	*Depth*	8	m	*Depth*	10	m	*Depth*	6	m
Thickness wall	1	m	Thickness wall	1	m	Thickness wall	1	m	Thickness wall	1	m
Total area cross-section	44	m2	*Total area cross-section*	28	m2	*Total area cross-section*	36	m2	*Total area cross-section*	20	m2
Total volume core	6600	m3	*Total volume core*	2100	m3	*Total volume core*	1800	m3	*Total volume core*	600	m3

Amount of cores	3	
Total volume cores combined	10500	m3
Reinforcement ratio	0.04	
Total volume concrete	10080	m3
Total volume steel	420	m3

Concrete floors for horizontal transfer forces between cores (assumption: core 2 and 3 are connected with main core; core 4 is connected with core 2 or 3)

Total dimensions tower 1			Total dimensions tower 2			Total dimensions tower 3			Total dimensions tower 4		
Width	30	m	*Width*	18	m	*Width*	18	m	*Width*	12	m
Length	30	m	*Length*	18	m	*Length*	18	m	*Length*	12	m
Area floors around core	756	m2	*Area floors around core*	260	m2	*Area floors around core*	224	m2	*Area floors around core*	108	m2
Thickness floors	120	mm									
Height between concrete floors	16	m									
No. of floors between core 1 and 2	4.6875		*No. of floors between core 1 and 3*	3.125		*No. of floors between core 2 and 4*	1.875				
Volume floors tower 1	425.25	m3	*Volume floors tower 2*	146.25	m3	*Volume floors tower 3*	84	m3	*Volume floors tower 4*	24.3	m3

Total volume floors combined	655.5	m3

Reinforcement ratio	0.04		
Total volume concrete	629.28	m3	
Total volume steel	26.22	m3	
Total system	11155.5	m3	
Total volume concrete	10709.28	m3	
Total volume steel	446.22	m3	
Embodied carbon concrete	**0.163**	**kg/kg**	
Embodied carbon steel	**NOT USED DIRECTLY**		
Modification factor			
Density steel	7850	kg/m3	
Weight steel	3502827	kg	total
	314	kg	per m3
Modification factor	0.24178	kg/kg	
Total embodied carbon	0.40478	kg/kg	
Density concrete	2300	kg/m3	
Weight concrete	24631344	kg	
Total embodied carbon	9970275	kg CO2	

REINFORCED CONCRETE - Modification Factors			
For reinforcement add this value to the appropriate concrete coefficient for each 100 kg of rebar per m3 of concrete	**1.04**	**0.077**	Add for each 100 kg steel rebar per m3 concrete. Use multiple of this value, i.e. for 150 kg steel use a factor of 1.5 times these values.

Timber trusses + CLT core

CLT Trusses

Grid size		
Width	5	m
Depth	5	m
Height story	4	m

Tower 1			*Tower 2*			*Tower 3*			*Tower 4*			*Tower 5*		
Height	150	m	Height	80	m	Height	60	m	Height	40	m	Height	30	m
No. of stories	37		*No. of stories*	20		*No. of stories*	15		*No. of stories*	10		*No. of stories*	7	
Dimensions			Dimensions			Dimensions			Dimensions			Dimensions		
Grid cells width	6		*Grid cells width*	4		*Grid cells width*	4		*Grid cells width*	3		*Grid cells width*	3	
Grid cells depth	6		*Grid cells depth*	4		*Grid cells depth*	4		*Grid cells depth*	3		*Grid cells depth*	3	

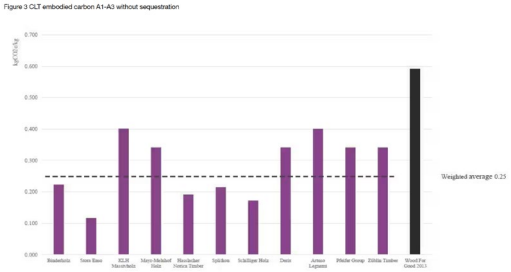
Dimensioning timber truss

Height	4	stories
Width	3	grid cells
Length truss	21.9317122	m

Dimensions truss

Width	1	m
Depth	1	m
Area truss	1	m2

Figure 3 CLT embodied carbon A1-A3 without sequestration



Total length truss

Tower 1	1228.175883	m
Tower 2	241.2488342	m
Tower 3	131.5902732	m
Tower 4	87.7268488	m
Tower 5	43.8634244	m
Total volume trusses	1732.605264	m3
Embodied carbon CO2	0.25	kg CO2/kg
Density clt	490	kg/m3
Total weight	848976.5792	kg
Total embodied carbon trusses	**212244.1448**	**kg**

CLT Cores

Main core			*Core 2*			*Core 3*			*Core 4*		
Height tower	150	m	Height tower	40	m	Height tower	50	m	Height tower	30	m
Dimensions			Dimensions			Dimensions			Dimensions		

Width	12	m	*Width*	8	m	*Width*	10	m	*Width*	6	m
Depth	12	m	*Depth*	8	m	*Depth*	10	m	*Depth*	6	m
Thickness wall	0.3	m	Thickness wall	0.3	m	Thickness wall	0.3	m	Thickness wall	0.3	m
Total area cross-section	14.04	m2	*Total area cross-section*	9.24	m2	*Total area cross-section*	11.64	m2	*Total area cross-section*	6.84	m2
Total volume core	2106	m3	*Total volume core*	369.6	m3	*Total volume core*	582	m3	*Total volume core*	205.2	m3

Amount of cores	1	
Total volume cores combined	2106	m3
Total weight cores	1031940	kg
Total embodied carbon clt cores	**257985**	**kg**
TOTAL EMBODIED CARBON	470229.1448	kg CO2

Timber trusses + Concrete core

CLT Trusses

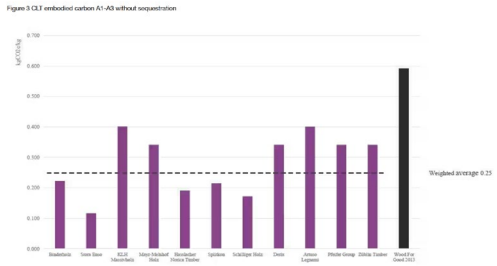
Grid size			
	Width	5	m
	Depth	5	m

Height story			
		4	m

Tower 1				*Tower 2*				*Tower 3*				*Tower 4*				*Tower 5*			
Height		150	m	Height		80	m	Height		60	m	Height		40	m	Height		30	m
	No. of stories	37			*No. of stories*	20			*No. of stories*	15			*No. of stories*	10			*No. of stories*	7	
Dimensions				Dimensions				Dimensions				Dimensions				Dimensions			
	Grid cells width	6			*Grid cells width*	4			*Grid cells width*	4			*Grid cells width*	3			*Grid cells width*	3	
	Grid cells depth	6			*Grid cells depth*	4			*Grid cells depth*	4			*Grid cells depth*	3			*Grid cells depth*	3	

Dimensioning timber truss

	Height	4	stories
	Width	3	grid cells
Length truss		21.9317122	m
Dimensions truss			
Width		1	m
Depth		1	m
	Area truss	1	m2



Total length truss		
Tower 1	1228.175883	m
Tower 2	241.2488342	m
Tower 3	131.5902732	m
Tower 4	87.7268488	m
Tower 5	43.8634244	m
Total volume trusses	1732.605264	m3

Embodied carbon CO2	0.25	kg CO2/kg
Density clt	490	kg/m3
Total weight	848976.5792	kg
Total embodied carbon trusses	**212244.1448**	**kg**

Concrete Cores

Main core		*Core 2*		*Core 3*		*Core 4*	
Height tower	150	Height tower	40 m	Height tower	50 m	Height tower	30 m
Dimensions		Dimensions		Dimensions		Dimensions	
Width	12 m	*Width*	8 m	*Width*	10 m	*Width*	6 m

Depth	12	m	*Depth*	8	m	*Depth*	10	m	*Depth*	6	m
Thickness wall	0.3	m	Thickness wall	0.3	m	Thickness wall	0.3	m	Thickness wall	0.3	m
Total area cross-section	14.04	m2	*Total area cross-section*	9.24	m2	*Total area cross-section*	11.64	m2	*Total area cross-section*	6.84	m2
Total volume core	2106	m3	*Total volume core*	369.6	m3	*Total volume core*	582	m3	*Total volume core*	205.2	m3

Amount of cores	1	
Total volume cores combined	2106	m3
Total weight cores	4843800	kg
Total embodied carbon concrete cores	**789539.4**	**kg**
TOTAL EMBODIED CARBON	1001783.545	kg CO2
Modification factor	0.24178	kg/kg
Embodied carbon concrete	**0.163**	**kg/kg**
Density concrete	2300	kg/m3

A.4 Imposed windloads

Building	A	B	C	D	E	F
Buildingheight [m]	152	72	56	72	40	24
Building width [facing wir	30	15	15	15	30	15
cr(z)	1,90839073	1,65896444	1,57507362	1,65896444	1,46275643	1,29223864
Mean wind speed	46,7555729	40,6446287	38,5893036	40,6446287	35,8375324	31,6598468
Turbulence intensity	0,17491607	0,2012148	0,21193181	0,2012148	0,22820491	0,25831777
Peak velocity pressure kN	3,03921976	2,48675867	2,31143682	2,48675867	2,08497475	1,75925761
Slenderness	5,06666667	4,8	3,73333333	4,8	1,33333333	1,6
cpe,windward	0,8	0,8	0,8	0,8	0,8	0,8
cpe, leeward	-0,6133333	-0,6	-0,5466667	-0,6	-0,4266667	-0,44
cf	1,41333333	1,4	1,34666667	1,4	1,22666667	1,24
cf,red	1,20133333	1,19	1,14466667	1,19	1,04266667	1,054
q,eq,peak	3,651116	2,95924281	2,64582468	2,95924281	2,17393367	1,85425752
q,avg,overtower	2,1906696	1,77554569	1,58749481	1,77554569	1,3043602	1,11255451

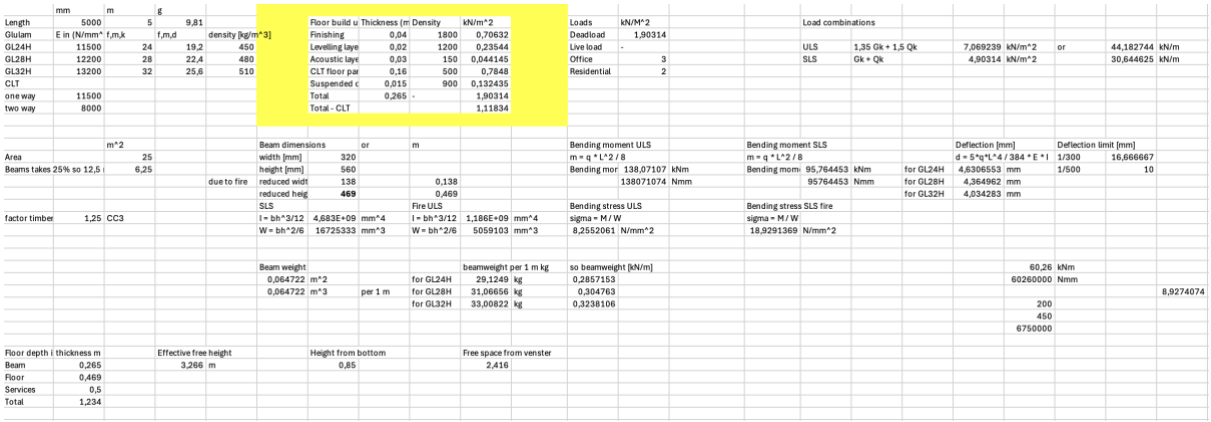
wind avg from north so towers A B C	2,05825791	kN/m^2
wind avg from west so towers A D E	1,69116125	kN/m^2

Wind area	3		Terrain categ	3		Air density	1,25	
Vb,0 [m/s]	24,5		z0	0,5				
c_dir	1		c0(z)	1				
C_season	1							
basic windspeed			Terrain factor					
vb = c_dir*c_season*vb,0			cr(z)	ln(h/z0) / ln(10/z0)				
vb	24,5		see row 3 under building					
mean wind speed at a height			Turbulence intensity					
vm(z) = cr(z) *c0 * vb			Iv(z) = 1/ ln (z/z0)					
Peak velocity pressure qp(z)			cpe, leeward					
qp(z) = (1+7 * Iv(z)) * 0,5 * p * vm^2			cpe = -0,5 + ((h/b - 2,8)/4) * -0,2					
cf,red.	0,85 * cf							
A factor of 0,6 is used to approximate the avg windpressure over the a whole tower for terrain III according to the eurocode								

A.5 Beam dimensioning

Beam dimensioning

Timber



	mm	m	g																					
Length	5000	5	9,81		Floor build u	Thickness (m	Density	kN/m^2		Loads	kN/M^2			Load combinations										
Glulam	E in (N/mm^	f,m,k	f,m,d	density [kg/m^3]	Finishing	0,04	1800	0,70632		Deadload	1,90314													
GL24H	11500	24	19,2	450	Levelling laye	0,02	1200	0,23544		Live load	-			ULS	1,35 Gk + 1,5 Qk		7,069239	kN/m^2	or	44,182744	kN/m			
GL28H	12200	28	22,4	480	Acoustic laye	0,03	150	0,044145		Office	3			SLS	Gk + Qk		4,90314	kN/m^2		30,644625	kN/m			
GL32H	13200	32	25,6	510	CLT floor par	0,16	500	0,7848		Residential	2													
CLT					Suspended c	0,015	900	0,132435																
one way	11500				Total	0,265	-	1,90314																
two way	8000				Total - CLT			1,11834																
		m^2			Beam dimensions		or	m		Bending moment ULS				Bending moment SLS				Deflection [mm]			Deflection limit [mm]		
Area		25			width [mm]	320				m = q * L^2 / 8				m = q * L^2 / 8				d = 5*q*L^4 / 384 * E * I		1/300	16,666667		
Beams takes 25% so	12,5	6,25			height [mm]	560				Bending mor	138,07107	kNm		Bending mom	95,764453	kNm	for GL24H	4,6306553	mm	1/500	10		
				due to fire	reduced widt	138		0,138			138071074	Nmm			95764453	Nmm	for GL28H	4,364962	mm				
					reduced heig	**469**		0,469									for GL32H	4,034283	mm				
					SLS			Fire ULS		Bending stress ULS				Bending stress SLS fire										
factor timber	1,25	CC3			I = bh^3/12	4,683E+09	mm^4	I = bh^3/12	1,186E+09	mm^4	sigma = M / W			sigma = M / W										
					W = bh^2/6	16725333	mm^3	W = bh^2/6	5059103	mm^3	8,2552061	N/mm^2		18,9291369	N/mm^2									
					Beam weight				beamweight per 1 m kg		so beamweight [kN/m]								60,26	kNm				
					0,064722	m^2		for GL24H	29,1249	kg	0,2857153								60260000	Nmm				
					0,064722	m^3	per 1 m	for GL28H	31,06656	kg	0,304763													8,9274074
								for GL32H	33,00822	kg	0,3238106								200					
																			450					
																			6750000					
Floor depth (	thickness m		Effective free height			Height from bottom			Free space from venster															
Beam	0,265		3,266	m		0,85			2,416															
Floor	0,469																							
Services	0,5																							
Total	1,234																							

Concrete dimensioning

			g											
			9,81								Roof buildup	Thickness [m	Desntiy [kN/m	Deadlaod [kN/m^2]
Strength class						meter	millimeter				Paving finish	0,06	24	1,44
Concrete cla	Elastic modu	F_ck	F_cd	density [kN/r	Length						Drainage laye	0,05	18	0,9
C25/30	31000	25	14,166667	2400		7,5	7500				Protection la	0,005	2	0,01
C30/37	33000	30	17	2400		10	10000				Waterproofir	0,005	11	0,055
C35/45	34000	35	19,833333	2400		15	15000				Thermal insu	0,12	3	0,36
C40/50	35000	40	22,666667	2400							Vapour barrie	0,0005	9	0,0045
C45/55	36000	45	25,5	2400							CLT panel	0,14	4,8	0,672
C50/60	37000	50	28,333333	2400							Ceilling	0,015	8,5	0,1275
											Total	0,3955		3,569
gamma factc	1,5													

Loads	kN/M^2		Load combinations						
Deadload	3,569								
Live load	-	ULS	1,35 Gk + 1,5 Qk + 1,1 Sk	8,61015	kN/m^2	or	121,0802	kN/m	
Terrace (reside	2	SLS	Gk + Qk	5,569	kN/m^2		78,31406	kN/m	
Snow	0,72	SLS (snow)	Gk +0,3 Qk + 0,2Sk	4,313	kN/m^2		60,65156	kN/m	

15 x 15 grid

Beam dimen	15 m (in mm)	in [m]	7,5	in [m]	Bending mor	15 m	7,5 m	
width [mm]	600	0,6	270	0,27	m = q * L^2 / 8			
height [mm]	1000	1	1200	1,2	Bending mor	3405,3816	851,345398	kNm
						3,405E+09	851345398	Nmm
I = bh^3/12	5E+10	mm^4	3,888E+10					
W = bh^2/6	100000000	mm^3	64800000		Bending stre	15 m	7,5 m	
					sigma = M / W			
						34,053816	13,1380463	

7.5 x 7.5 grid

For 15 m	Deflection [mm]	Deflection limit [mm]		For 7,5 m	Deflection [mm]	Deflection limit [mm]	
	d = 5*q*L^4 / 384 * E * I	1/300	50		d = 5*q*L^4 / 384 * E * I	1/300	25
C25/30	33,305187	1/500	30	C25/30	2,676922	1/500	15
C30/37	31,28669			C30/37	2,514684		
C35/45	30,366494			C35/45	2,440723		
C40/50	29,49888			C40/50	2,370988		
C45/55	28,679466			C45/55	2,305127		
C50/60	27,904346			C50/60	2,242826		

A.6 Column dimensioning

Timber dimensioning

Zones	Floors					
1		7	Deadload			
2	2 to 8	7	floor	1,90314	kN/m^2	
3	9 to 16	7	Beam			
4	17 - 24	7	for GL24H	0,2857153	kN/m	
5	25 - 32	6	for GL28H	0,304763	kN/m	
6	33 - 38		for GL32H	0,3238106	kN/m	
beamlength	5		Liveload	3	kN/m^2	
column supports						
4*L/2						
10	m					
floor area per colmn						
25	m^2					

Deadload			ULS	1,35 Gk + 1,5 Qk			180,58813	kN	now GL24H				
Floor	47,5785	kN	SLS	Gk + Qk			125,43565	kN					
Beam													
for GL24H	2,8571527	kN											
for GL28H	3,0476295	kN									Nreduced = n*1,35*Gk + phi * n *1,5* Qk		
for GL32H	3,2381064	kN									Load per zone from floor		
Total dead						zones	floors per zone				ULS	SLS	
for GL24H	50,435653	kN				1	4	0,4			454,40967	321,74261	kN
for GL28H	50,62613	kN				2	7	0,4			795,21693	563,04957	kN
for GL32H	50,816606	kN				3	7	0,4			795,21693	563,04957	kN
						4	7	0,4			795,21693	563,04957	kN
Liveload	75	kN				5	7	0,4			795,21693	563,04957	kN
						6	6	0,4			681,61451	482,61392	kN
							38						
			based on A = N/f										

					based on A = N/f					
					Estimated A mm					
Timber type	f,c,0,k [N/mr	E-mod	gamma	f,c,0,d	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
C24	21	11000	1,25	16,8	205566,28	183927,73	146060,25	108192,78	70325,307	32457,834
C30	24	12000	1,25	19,2	179870,5	160936,76	127802,72	94668,682	61534,643	28400,605
C35	25	13000	1,25	20	172675,68	154499,29	122690,61	90881,935	59073,258	27264,58
C40	26	14000	1,25	20,8	166034,3	148557,01	117971,74	87386,476	56801,209	26215,943
GL24H	24	11000	1,25	19,2	179870,5	160936,76	127802,72	94668,682	61534,643	28400,605
GL28H	26,5	12000	1,25	21,2	162901,58	145754,05	115745,86	85737,674	55729,488	25721,302
Gl32H	29	13000	1,25	23,2	148858,34	133189,04	105767,77	78346,496	50925,222	23503,949
GL36H	30	14000	1,25	24	143896,4	128749,41	102242,18	75734,946	49227,715	22720,484

	Estimated width and height mm^2					
Timber type	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
C24	453,39418	428,86796	382,1783	328,92671	265,18919	180,16058
C30	424,11142	401,16924	357,49506	307,68276	248,06177	168,52479
C35	415,54263	393,06398	350,2722	301,46631	243,04991	165,1199
C40	407,47307	385,43094	343,47015	295,61204	238,33004	161,91338
GL24H	424,11142	401,16924	357,49506	307,68276	248,06177	168,52479
GL28H	403,61068	381,77748	340,21443	292,80996	236,07094	160,37862
Gl32H	385,82164	364,95074	325,21957	279,90444	225,66617	153,30998
GL36H	379,33679	358,81668	319,75331	275,19983	221,8732	150,73315

Buckling check timber

	Dimensioning inculding reduction due to fire safety										
	Verdiepingen		7	7	7	7	6				
		Chosen dimensions		Chosen material		GI32H					
ULS		Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6				
	witdh [mm]	720	720	720	640	600	520				
	height [mm]	720	720	720	640	600	520				
	area [mm^2]	518400	518400	518400	409600	360000	270400		GL32H		
	I = bh^3/12	22394880000	22394880000	22394880000	13981013333	10800000000	6093013333		sqrt Fc/E0	0,0422447	
	i = sqrt(I/A)	207,8460969	207,8460969	207,8460969	184,7520861	173,2050808	150,11107		lamda = leff/i * sqrt fc/E0mean		
	slenderness	0,812999791	0,812999791	0,812999791	0,914624765	0,97559975	1,125692019				
	ϕ	0,88178431	0,88178431	0,88178431	0,979731707	1,043457411	1,216160463				
	x	0,817517779	0,817517779	0,817517779	0,75135792	0,707411694	0,5964996				
	Nrd	8984,585795	8136,983361	8136,983361	5908,919118	4889,629632	3096,835046				
	UC-ULS	0,480477565	0,474682331	0,376953616	0,249932587	0,302033396	0,220100361				
Fire safety SLS											
	reduced width [r	538	538	538	458	418	338				
	reduced height [	538	538	538	458	418	338				
	reduced Area [m	289444	289444	289444	209764	174724	114244				
	I = bh^3/12	6981485761	6981485761	6981485761	3666744641	2544039681	1087640961				
	i = sqrt(I/A)	155,3072224	155,3072224	155,3072224	132,2132116	120,6662063	97,57219549				
	slenderness	1,088029461	1,088029461	1,088029461	1,278078275	1,400382416	1,731833875				
	ϕ	1,170707	1,170707	1,170707	1,414549866	1,590573697	2,142807673				
	x	0,623889059	0,623889059	0,623889059	0,494868062	0,426476033	0,2937115			999,488	
	Nrd	3467,154139	3467,154139	3467,154139	1993,065678	1430,699488	644,2517114				
	UC-SLS	0,881574536	0,788777217	0,626381907	0,807155063	0,730875697	0,749107697				

Concrete dimensioning

				15 x 15 m						15 x 15 m					
Loads				Axial load per column						Axial load per column					
				Deadload			Deadload	Total		Deadload			Deadload	Total	
Deadload				roof,1	602,26875	kN	1	1026,0608		roof,1	200,75625	kN	1	315,18009	
roof	3,569	kN/m^2		roof,2	401,5125	kN	2	719,3565		roof,2	150,567188	kN	2	236,38507	
beam	15 x 15	7,5 x 7,5		roof,3	200,75625	kN	3	412,65225		roof,3	100,378125	kN	3	157,59005	
C25/30	14,1264	7,628256	kN/m												
C30/37	14,1264	7,628256	kN/m	beam,1	423,792	kN	Liveload	Total		beam,1	114,42384	kN	Liveload	Total	
C35/45	14,1264	7,628256	kN/m	beam,2	317,844	kN	1	459		beam,2	85,81788	kN	1	153	
C40/50	14,1264	7,628256	kN/m	beam,3	211,896	kN	2	306		beam,3	57,21192	kN	2	114,75	
C45/55	14,1264	7,628256	kN/m				3	153					3	76,5	
C50/60	14,1264	7,628256	kN/m	Liveload						Liveload					
				Terrace,1	337,5	kN				Terrace,1	112,5	kN			
Liveload				Terrace,2	225	kN				Terrace,2	84,375	kN			
Terrace	2	kN/m^2		Terrace,3	112,5	kN				Terrace,3	56,25	kN			
Snow	0,72	kN/m^2		Snow,1	121,5	kN				Snow,1	40,5	kN			
				Snow,2	81	kN				Snow,2	30,375	kN			
				Snow,3	40,5	kN				Snow,3	20,25	kN			

Some columns will have a different load from the floor which they need to carry					
Floor area per column		15 x 15 m	Floor area per	7,5 x 7,5 m	
1	168,75	m^2	1	56,25	m^2
2	112,5	m^2	2	42,1875	m^2
3	56,25	m^2	3	28,125	m^2
Beam length per column			Beam length per column		
1	30	m	1	15	m
2	22,5	m	2	11,25	m
3	15	m	3	7,5	m
currently we assume that 15 x 15 grid					
maybe 15 x 7,5 grid is better					

15 x 15 grid

	Load combinations						Column lengt	8	m							
								8000	mm							
	ULS	1,35 Gk + 1,5 Qk + 1,1 Sk					leff	6,8	m	fixed pinned is assumed						
	SLS	Gk + Qk						6800	mm							
	SLS (snow)	Gk + 0,3 Qk + 0,2Sk														
	15 x 15 m															
	Column 1			Column 2			Column 3									
	ULS	2025,082	kN	ULS	1397,7313	kN	ULS	770,38054	kN							
	SLS	1363,5608	kN	SLS	944,3565	kN	SLS	525,15225	kN							
	SLS(snow)	1151,6108	kN	SLS(snow)	803,0565	kN	SLS(snow)	454,50225	kN							
	Gamma	1,5														
				Stress				Buckling								
Column 1	Type of concr	Elastic modu	F_ck	F_cd	Area[mm^2]	width [mm]		Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60		
	C25/30	31000	25	14,166667	142946,97	378,08328		width [mm]	400	350	320	300	280	270		
	C30/37	33000	30	17	119122,47	345,14123		height [mm]	400	350	320	300	280	270		
	C35/45	34000	35	19,833333	102104,98	319,53869		Area[mm^2]	160000	122500	102400	90000	78400	72900		
	C40/50	35000	40	22,666667	89341,853	298,90108		I [mm^4]	2,133E+09	1,251E+09	873813333	675000000	512213333	442867500		
	C45/55	36000	45	25,5	79414,981	281,80664		i [mm]	115,47005	101,0363	92,3760431	86,60254	80,829038	77,942286		
	C50/60	37000	50	28,333333	71473,483	267,34525		lambda	58,889727	67,302546	73,6121593	78,519637	84,128182	87,244041	slenderness	lambda < 90
								Nb (euler)	14115,697	8808,1924	6341,31759	5042,5909	3935,8207	3497,4978		
								UC	0,1434631	0,2299089	0,3193472	0,4015955	0,514526	0,5790088		
Column 2	Type of concr	Elastic modu	F_ck	F_cd	Area[mm^2]	width [mm]		Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60		
	C25/30	31000	25	14,166667	98663,384	314,10728		width [mm]	320	300	270	250	240	230		
	C30/37	33000	30	17	82219,487	286,73941		height [mm]	320	300	270	250	240	230		
	C35/45	34000	35	19,833333	70473,846	265,46911		Area[mm^2]	102400	90000	72900	62500	57600	52900		
	C40/50	35000	40	22,666667	61664,615	248,32361		I [mm^4]	873813333	675000000	442867500	325520833	276480000	233200833		
	C45/55	36000	45	25,5	54812,991	234,12174		i [mm]	92,376043	86,60254	77,9422863	72,168784	69,282032	66,395281		
	C50/60	37000	50	28,333333	49331,692	222,10739		lambda	73,612159	78,519637	87,2440407	94,223564	98,149546	102,41692		
								Nb (euler)	5781,7896	4754,4429	3213,91693	2431,805	2124,458	1841,6782		
								UC	0,2417472	0,2939842	0,43489963	0,5747711	0,6579237	0,7589444		
Column 3	Type of concr	Elastic modu	F_ck	F_cd	Area[mm^2]	width [mm]		Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60		
	C25/30	31000	25	14,166667	54379,803	233,19477		width [mm]	240	220	200	200	200	200		
	C30/37	33000	30	17	45316,502	212,87673		height [mm]	240	220	200	200	200	200		
	C35/45	34000	35	19,833333	38842,716	197,08556		Area[mm^2]	57600	48400	40000	40000	40000	40000		
	C40/50	35000	40	22,666667	33987,377	184,35666		I [mm^4]	276480000	195213333	133333333	133333333	133333333	133333333		
	C45/55	36000	45	25,5	30211,001	173,81312		i [mm]	69,282032	63,50853	57,7350269	57,735027	57,735027	57,735027		
	C50/60	37000	50	28,333333	27189,901	164,89361		lambda	98,149546	107,07223	117,779455	117,77945	117,77945	117,77945		
								Nb (euler)	1829,3944	1375,0084	967,608275	996,06734	1024,5264	1052,9855		
								UC	0,4211123	0,5602734	0,79616985	0,7734221	0,7519382	0,7316155		

7.5 x 7.5 grid

	7,5 x 7,5 m															
	Column 1			Column 2			Column 3									
	ULS	638,793122	kN	ULS	479,0948411	kN	ULS	319,39656								
	SLS	427,68009	kN	SLS	320,7600675	kN	SLS	213,84005								
	SLS(snow)	357,03009	kN	SLS(snow)	267,7725675	kN	SLS(snow)	178,51505								
	Gamma	1,5														
Column 1	Type of concrete	Elastic modulu	F_ck	F_cd	Area[mm^2]	width [mm]		Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60		
	C25/30	31000	25	14,166667	45091,27916	212,34707		width [mm]	400	350	320	300	280	270		
	C30/37	33000	30	17	37576,06597	193,84547		height [mm]	400	350	320	300	280	270		
	C35/45	34000	35	19,833333	32208,05655	179,46603		Area[mm^2]	160000	122500	102400	90000	78400	72900		
	C40/50	35000	40	22,666667	28182,04948	167,8751		I [mm^4]	2,133E+09	1,251E+09	873813333	675000000	512213333	442867500		
	C45/55	36000	45	25,5	25050,71065	158,27416		i [mm]	115,47005	101,0363	92,376043	86,60254	80,829038	77,942286		
	C50/60	37000	50	28,333333	22545,63958	150,15205		lambda	58,889727	67,302546	73,612159	78,519637	84,128182	87,244041	slenderness	lambda < 90
								Nb (euler)	14115,697	8808,1924	6341,3176	5042,5909	3935,8207	3497,4978		
								UC	0,1434631	0,2299089	0,3193472	0,4015955	0,514526	0,5790088		
Column 2	Type of concrete	Elastic modulu	F_ck	F_cd	Area[mm^2]	width [mm]		Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60		
	C25/30	31000	25	14,166667	33818,45937	183,89796		width [mm]	320	300	270	250	240	230		
	C30/37	33000	30	17	28182,04948	167,8751		height [mm]	320	300	270	250	240	230		
	C35/45	34000	35	19,833333	24156,04241	155,42214		Area[mm^2]	102400	90000	72900	62500	57600	52900		
	C40/50	35000	40	22,666667	21136,53711	145,3841		I [mm^4]	873813333	675000000	442867500	325520833	276480000	233200833		
	C45/55	36000	45	25,5	18788,03299	137,06945		i [mm]	92,376043	86,60254	77,942286	72,168784	69,282032	66,395281		
	C50/60	37000	50	28,333333	16909,22969	130,03549		lambda	73,612159	78,519637	87,244041	94,223564	98,149546	102,41692		
								Nb (euler)	5781,7896	4754,4429	3213,9169	2431,805	2124,458	1841,6782		
								UC	0,2417472	0,2939842	0,4348996	0,5747711	0,6579237	0,7589444		
Column 3	Type of concrete	Elastic modulu	F_ck	F_cd	Area[mm^2]	width [mm]		Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60		
	C25/30	31000	25	14,166667	22545,63958	150,15205		width [mm]	240	220	200	200	200	200		
	C30/37	33000	30	17	18788,03299	137,06945		height [mm]	240	220	200	200	200	200		
	C35/45	34000	35	19,833333	16104,02827	126,90165		Area[mm^2]	57600	48400	40000	40000	40000	40000		
	C40/50	35000	40	22,666667	14091,02474	118,70562		I [mm^4]	276480000	195213333	133333333	133333333	133333333	133333333		
	C45/55	36000	45	25,5	12525,35532	111,91673		i [mm]	69,282032	63,50853	57,735027	57,735027	57,735027	57,735027		
	C50/60	37000	50	28,333333	11272,81979	106,17354		lambda	98,149546	107,07223	117,77945	117,77945	117,77945	117,77945		
								Nb (euler)	1829,3944	1375,0084	967,60827	996,06734	1024,5264	1052,9855		
								UC	0,4211123	0,5602734	0,7961699	0,7734221	0,7519382	0,7316155		

A.7 Core dimensioning

Core dimensioning

	The 10 x 10 core goes upto 152 meter						Fire safety						
	The core replaces 9 columns						R120						
							tfire [min]	120	min				
	CLT types	F,ck [MPa]	gamma	F,cd [MPa]	E-mod [MPa]		effective cha	0,65	mm/min				
	C24	21	1,3	16,1538462	11000		1 side	85	mm				
	C30	24	1,3	18,4615385	12000		2 side	170	mm				
	C35	25	1,3	19,2307692	13000								
	GL24H	24	1,3	18,4615385	11000								
	GL28H	28	1,3	21,5384615	12000								
	GL32H	32	1,3	24,6153846	13000								
For core A1					A1 dimensions								
	ULS	cumulative p	per 9 columns		Grid	10 x 10	meters						
	Zone 1	3056,5548	27508,9932	kN	Number of wa	4							
	Zone 2	2734,81219	24613,3097	kN	Wall length [r	10	m						
	Zone 3	2171,76262	19545,8636	kN	Wall length [r	10000	mm						
	Zone 4	1608,71305	14478,4175	kN	Total wall len	40000	mm						
	Zone 5	1045,66348	9410,97136	kN									
	Zone 6	482,613916	4343,52525	kN									
	Area required per zone			Thickness per zone		Adding firesafety		Final thickenss		Ply buildup			
	Zone 1	1702937,68	mm^2	42,5734419		212,573442	mm	220	7-ply	40	24 x 5	40	mm
	Zone 2	1523681,08	mm^3	38,092027		208,092027	mm	220	7-ply	40	24 x 5	40	mm
	Zone 3	1209982,03	mm^4	30,2495508		200,249551	mm	200	7-ply	40	24 x 5	40	mm
	Zone 4	896282,987	mm^5	22,4070747		192,407075	mm	200	7-ply	40	24 x 5	40	mm
	Zone 5	582583,942	mm^6	14,5645985		184,564599	mm	200	7-ply	40	20 x 5	40	mm
	Zone 6	268884,896	mm^7	6,7221224		176,722122	mm	180	7-ply	40	20 x 5	40	mm

A.8 Unity checks

Columns

Timber compressive

Zone	Axial force N	Area of column	Stress on column	Allowed stress	UC
1	4316891,907	518400	8,327337784	19,2	0,43371551
2	3862482,233	518400	7,450775912	19,2	0,388061245
3	3067265,303	518400	5,916792636	19,2	0,308166283
4	2272048,372	409600	5,546993096	19,2	0,28890589
5	1476831,442	360000	4,102309561	19,2	0,213661956
6	681614,5117	270400	2,520763727	19,2	0,131289777

Timber buckling

	Dimensioning inculding reduction due to fire safety						
	Verdiepingen		7	7	7	7	6
		Chosen dimensions		Chosen material		GI32H	
ULS		Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
	witdh [mm]	720	720	720	640	600	520
	height [mm]	720	720	720	640	600	520
	area [mm^2]	518400	518400	518400	409600	360000	270400
	I = bh^3/12	22394880000	22394880000	22394880000	13981013333	10800000000	6093013333
	i = sqrt(I/A)	207,8460969	207,8460969	207,8460969	184,7520861	173,2050808	150,11107
	slenderness	0,812999791	0,812999791	0,812999791	0,914624765	0,97559975	1,125692019
	ϕ	0,88178431	0,88178431	0,88178431	0,979731707	1,043457411	1,216160463
	x	0,817517779	0,817517779	0,817517779	0,75135792	0,707411694	0,5964996
	Nrd	8984,585795	8136,983361	8136,983361	5908,919118	4889,629632	3096,835046
	UC-ULS	0,480477565	0,474682331	0,376953616	0,249932587	0,302033396	0,220100361
Fire safety SLS							
	reduced width [r	538	538	538	458	418	338
	reduced height [	538	538	538	458	418	338
	reduced Area [m	289444	289444	289444	209764	174724	114244
	I = bh^3/12	6981485761	6981485761	6981485761	3666744641	2544039681	1087640961
	i = sqrt(I/A)	155,3072224	155,3072224	155,3072224	132,2132116	120,6662063	97,57219549
	slenderness	1,088029461	1,088029461	1,088029461	1,278078275	1,400382416	1,731833875
	ϕ	1,170707	1,170707	1,170707	1,414549866	1,590573697	2,142807673
	x	0,623889059	0,623889059	0,623889059	0,494868062	0,426476033	0,2937115
	Nrd	3467,154139	3467,154139	3467,154139	1993,065678	1430,699488	644,2517114
	UC-SLS	0,881574536	0,788777217	0,626381907	0,807155063	0,730875697	0,749107697

Concrete compressive

Axial force k N	Axial force N	Area of colum	Stress on colum	Allowed stress	UC
638,7931215	638793,12	72900	8,76259426	28,33	0,309304421
479,0948411	479094,84	72900	6,57194569	28,33	0,231978316
319,3965608	319396,56	72900	4,38129713	28,33	0,154652211

Concrete buckling

Column 1	Type of concrete	Elastic modulus	F_ck	F_cd	Area [mm^2]	width [mm]
	C25/30	31000	25	14,1666667	45091,27916	212,347072
	C30/37	33000	30	17	37576,06597	193,845469
	C35/45	34000	35	19,8333333	32208,05655	179,466032
	C40/50	35000	40	22,6666667	28182,04948	167,875101
	C45/55	36000	45	25,5	25050,71065	158,274163
	C50/60	37000	50	28,3333333	22545,63958	150,152055

Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
width [mm]	400	350	320	300	280	270
height [mm]	400	350	320	300	280	270
Area [mm^2]	160000	122500	102400	90000	78400	72900
I [mm^4]	2,133E+09	1,251E+09	873813333	675000000	512213333	442867500
i [mm]	115,47005	101,0363	92,376043	86,60254	80,829038	77,942286
lambda	58,889727	67,302546	73,612159	78,519637	84,128182	87,244041
Nb (euler)	14115,697	8808,1924	6341,3176	5042,5909	3935,8207	3497,4978
UC	0,1434631	0,2299089	0,3193472	0,4015955	0,514526	0,5790088

Column 2	Type of concrete	Elastic modulus	F_ck	F_cd	Area [mm^2]	width [mm]
	C25/30	31000	25	14,1666667	33818,45937	183,897959
	C30/37	33000	30	17	28182,04948	167,875101
	C35/45	34000	35	19,8333333	24156,04241	155,422143
	C40/50	35000	40	22,6666667	21136,53711	145,384102
	C45/55	36000	45	25,5	18788,03299	137,069446
	C50/60	37000	50	28,3333333	16909,22969	130,035494

Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
width [mm]	320	300	270	250	240	270
height [mm]	320	300	270	250	240	270
Area [mm^2]	102400	90000	72900	62500	57600	72900
I [mm^4]	873813333	675000000	442867500	325520833	276480000	442867500
i [mm]	92,376043	86,60254	77,942286	72,168784	69,282032	77,942286
lambda	73,612159	78,519637	87,244041	94,223564	98,149546	87,244041
Nb (euler)	5781,7896	4754,4429	3213,9169	2431,805	2124,458	3497,4978
UC	0,2417472	0,2939842	0,4348996	0,5747711	0,6579237	0,3996375

Column 3	Type of concrete	Elastic modulus	F_ck	F_cd	Area [mm^2]	width [mm]
	C25/30	31000	25	14,1666667	22545,63958	150,152055
	C30/37	33000	30	17	18788,03299	137,069446
	C35/45	34000	35	19,8333333	16104,02827	126,901648
	C40/50	35000	40	22,6666667	14091,02474	118,705622
	C45/55	36000	45	25,5	12525,35532	111,916734
	C50/60	37000	50	28,3333333	11272,81979	106,173536

Concrete	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
width [mm]	240	220	200	200	200	270
height [mm]	240	220	200	200	200	270
Area [mm^2]	57600	48400	40000	40000	40000	72900
I [mm^4]	276480000	195213333	133333333	133333333	133333333	442867500
i [mm]	69,282032	63,50853	57,735027	57,735027	57,735027	77,942286
lambda	98,149546	107,07223	117,77945	117,77945	117,77945	87,244041
Nb (euler)	1829,3944	1375,0084	967,60827	996,06734	1024,5264	3497,4978
UC	0,4211123	0,5602734	0,7961699	0,7734221	0,7519382	0,2202662

Beams

Timber deflection

Deflection [mm]		Deflection limit [mm]	
d = 5*q*L^4 / 384 * E * I		1/300	16,666667
4,6306553	mm	1/500	10
4,364962	mm		
4,034283	mm		

Timber bending

Beam dimensions		or	m			Bending moment ULS			Bending moment SLS		
width [mm]	320					m = q * L^2 / 8			m = q * L^2 / 8		
height [mm]	560					Bending mor	138,07107	kNm	Bending mom	95,764453	kNm
reduced widt	138		0,138				138071074	Nmm		95764453	Nmm
reduced heig	**469**		0,469								
SLS			Fire ULS			Bending stress ULS			Bending stress SLS fire		
I = bh^3/12	4,683E+09	mm^4	I = bh^3/12	1,186E+09	mm^4	sigma = M / W			sigma = M / W		
W = bh^2/6	16725333	mm^3	W = bh^2/6	5059103	mm^3	8,2552061	N/mm^2		18,9291369	N/mm^2	

Beam weight				beamweight per 1 m kg		so beamweight [kN/m]
0,064722	m^2		for GL24H	29,1249	kg	0,2857153
0,064722	m^3	per 1 m	for GL28H	31,06656	kg	0,304763
			for GL32H	33,00822	kg	0,3238106

Concrete deflection

For 7,5 m	Deflection [mm]		Deflection limit [mm]	
	d = 5*q*L^4 / 384 * E * I		1/300	25
C25/30	2,676922		1/500	15
C30/37	2,514684			
C35/45	2,440723			
C40/50	2,370988			
C45/55	2,305127			
C50/60	2,242826			

Concrete bending

Beam dimen	15 m (in mm)	in [m]	7,5	in [m]	Bending mo	15 m	7,5 m	
width [mm]	600	0,6	270	0,27	m = q * L^2 / 8			
height [mm]	1000	1	1200	1,2	Bending mo	3405,3816	851,345398	kNm
						3,405E+09	851345398	Nmm
I = bh^3/12	5E+10	mm^4	3,888E+10					
W = bh^2/6	100000000	mm^3	64800000		Bending stre	15 m	7,5 m	
					sigma = M / W			
						34,053816	13,1380463	

Combined unity checks

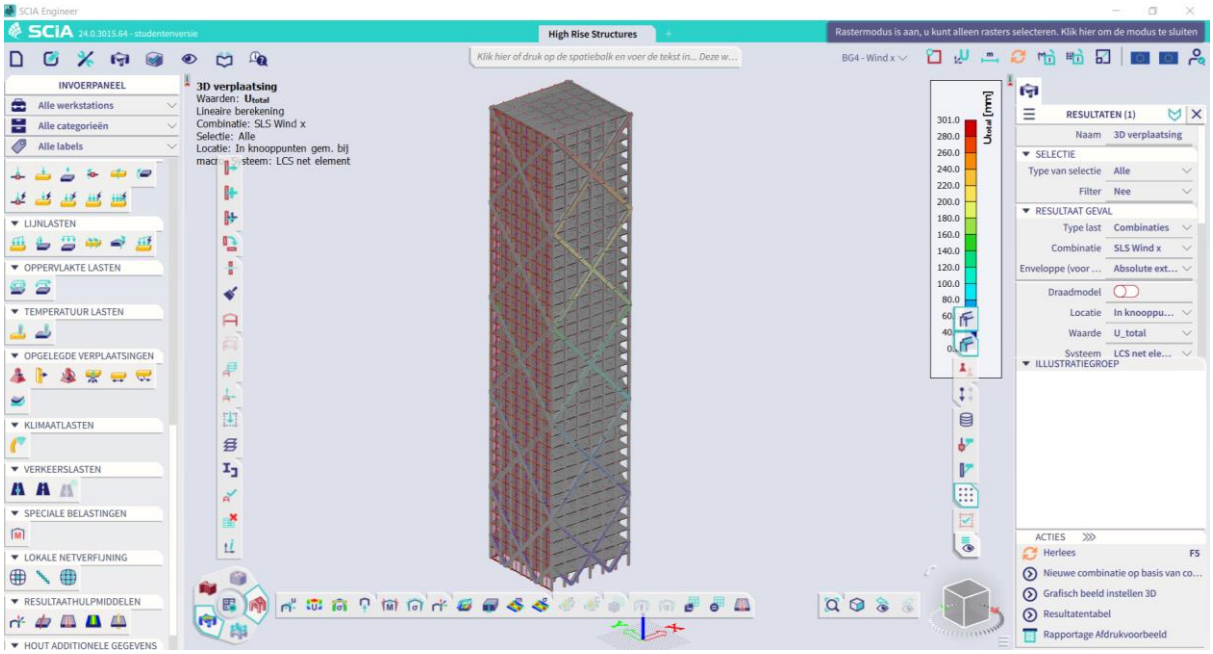
Deflection	Allowable	Actual	UC
C35/45	15	2,440723	0,162715
GL24H	10	4,630655	0,463066
Bending stres	Allowable	Actual	UC
C35/45	19,83333	13,13805	0,662423
GL24H	19,2	8,255206	0,429959

Cores axial

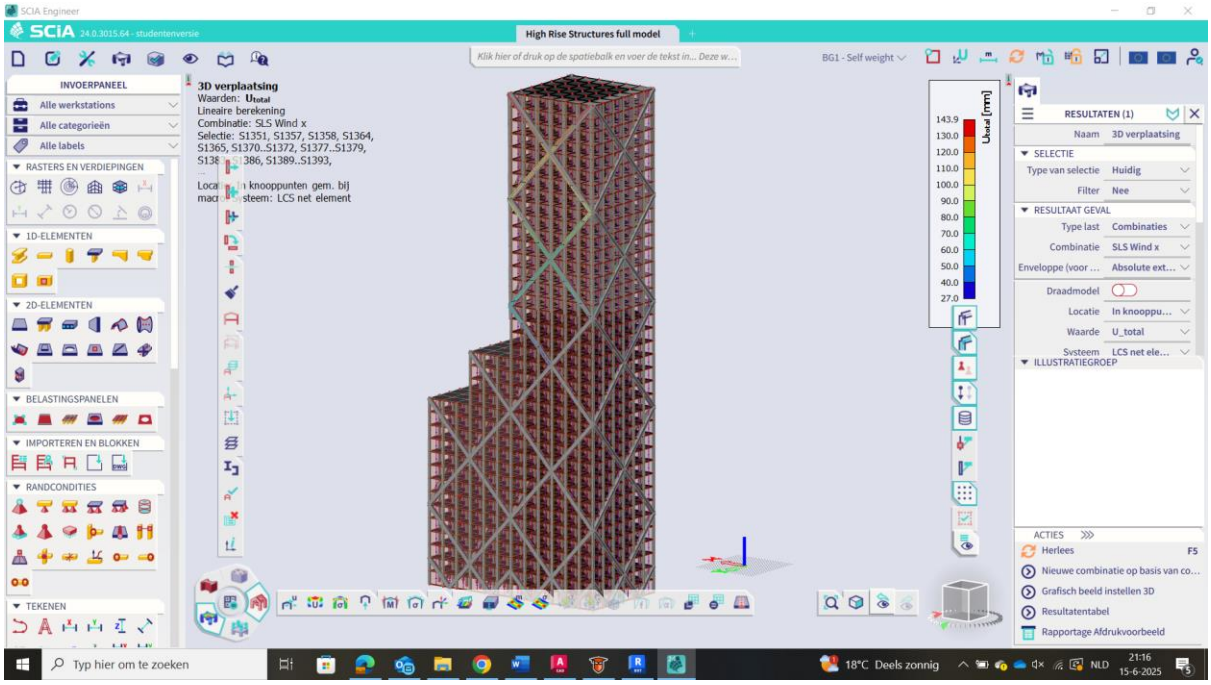
Zone	ll Thickness [m	Axial Load [kN]	Area [mm^2]	ompressive UC
1	220	27508,99	8800000	0,194
2	220	24613,31	8800000	0,173
3	200	19545,86	8000000	0,151
4	200	14478,42	8000000	0,112
5	200	9410,97	8000000	0,073
6	180	4343,53	7200000	0,037

A.9 SCIA Engineer results

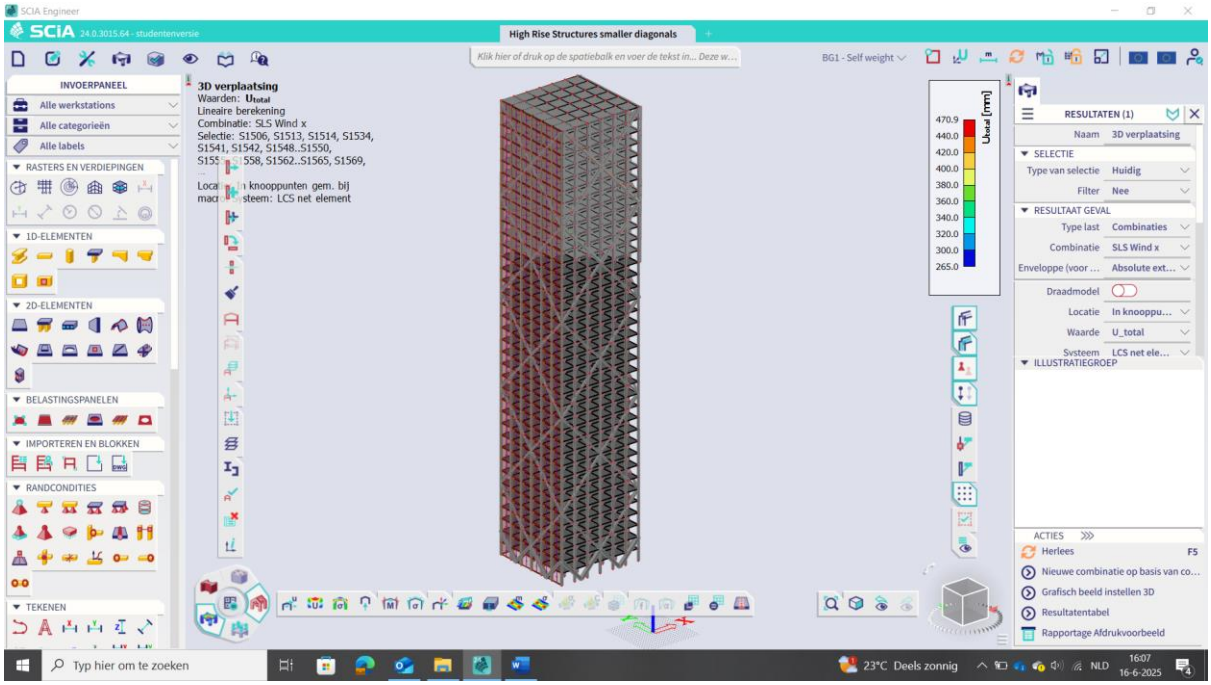
One core with all elements (diagonals 1.5x.1.5m)



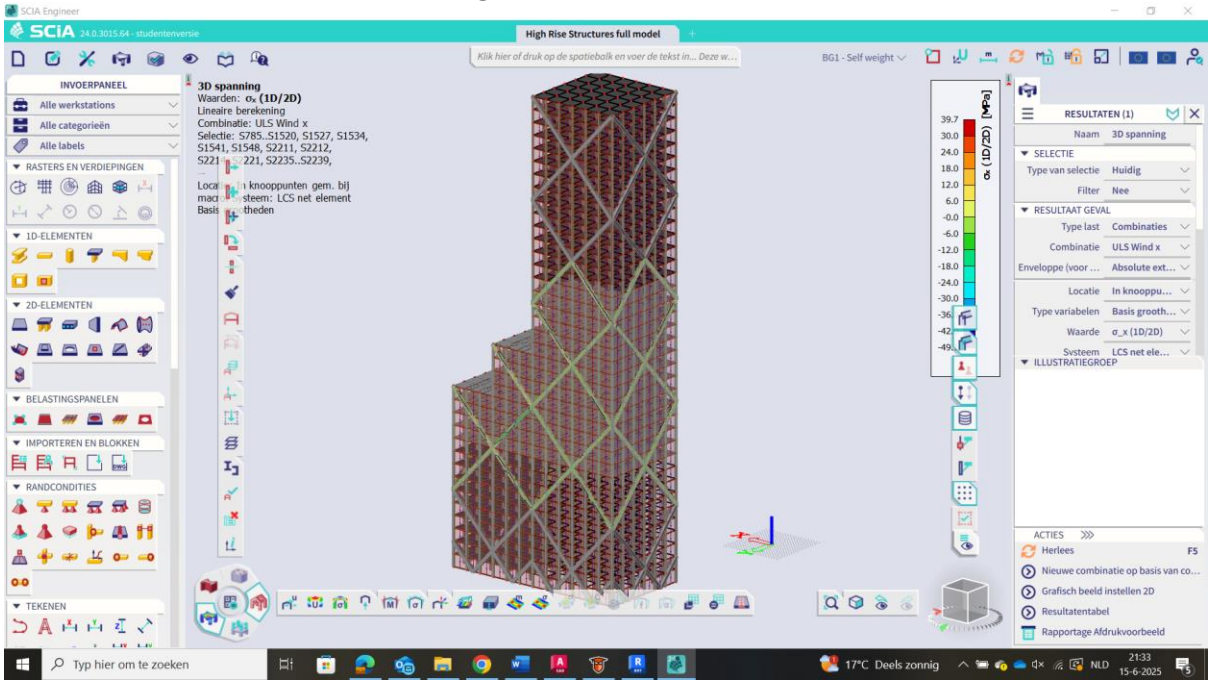
Three cores with all elements (diagonals 1.5x.1.5m)



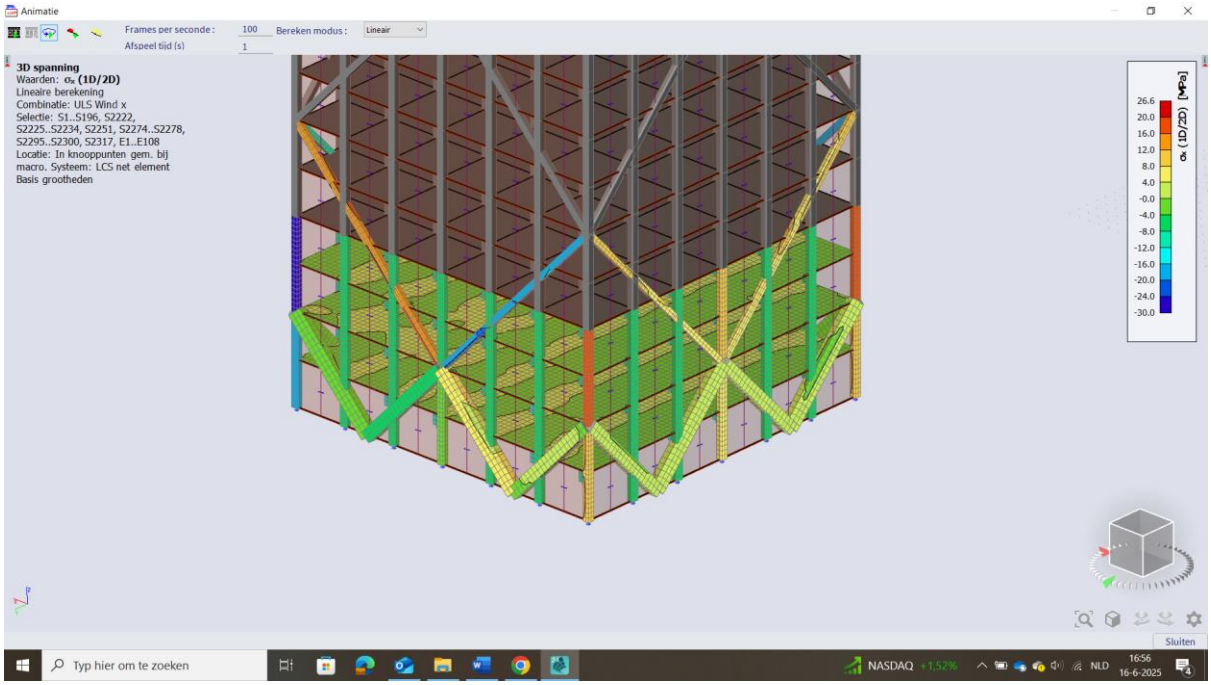
One core with all elements (diagonals 1.0x.1.0m)



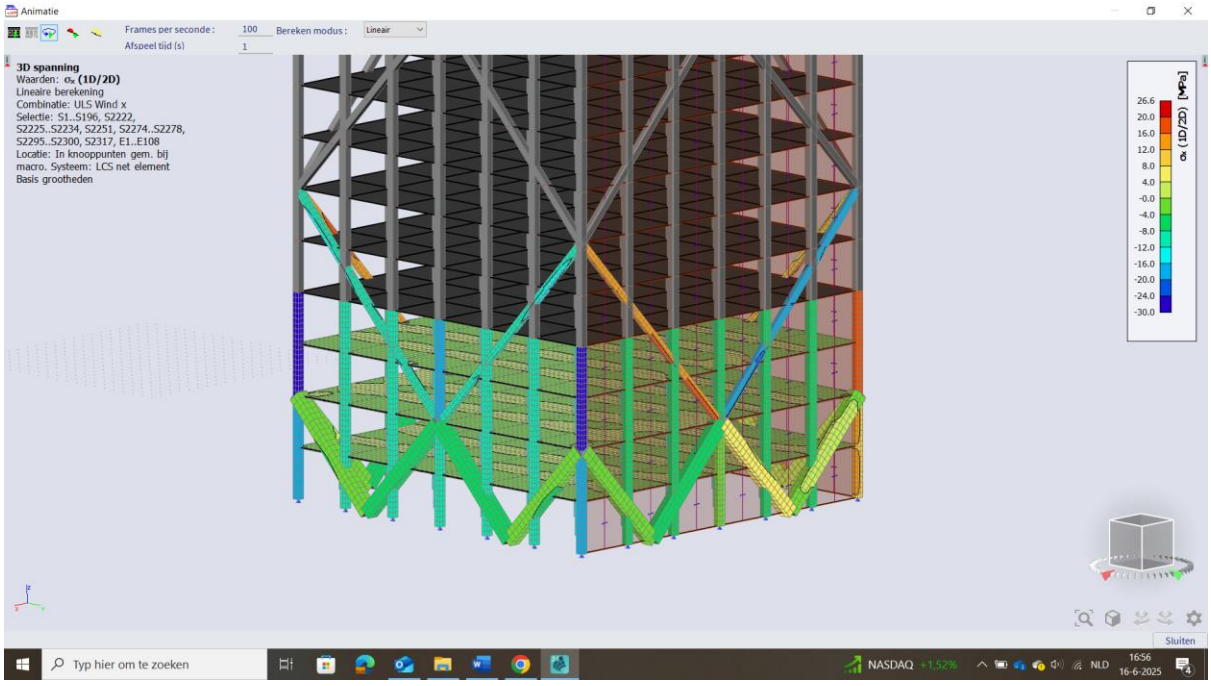
Three cores with all elements (diagonals 1.0x.1.0m)



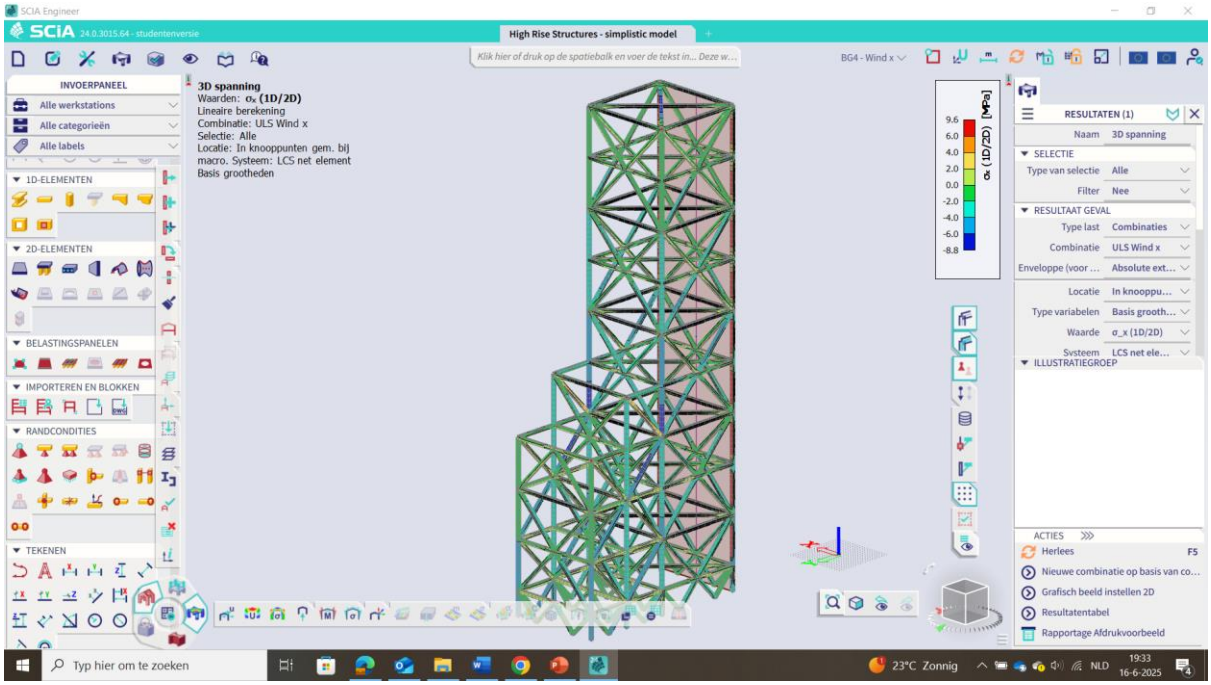
Stresses in bottom elements (south-west)



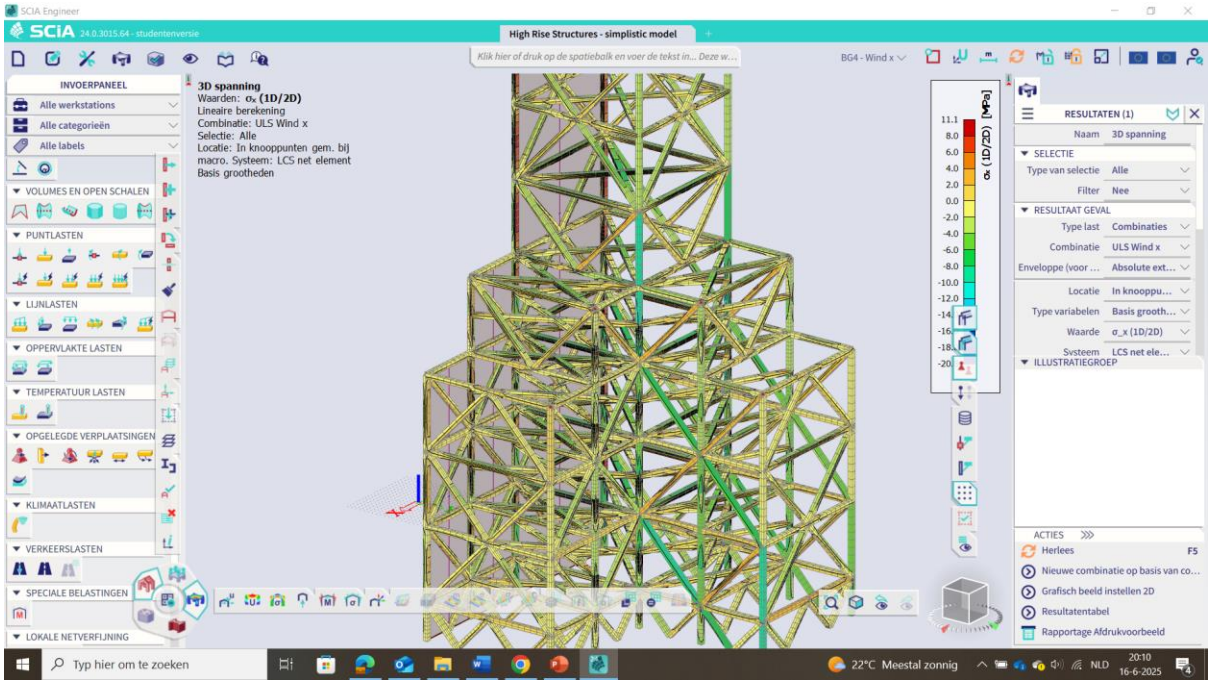
Stresses in bottom elements (north-east)



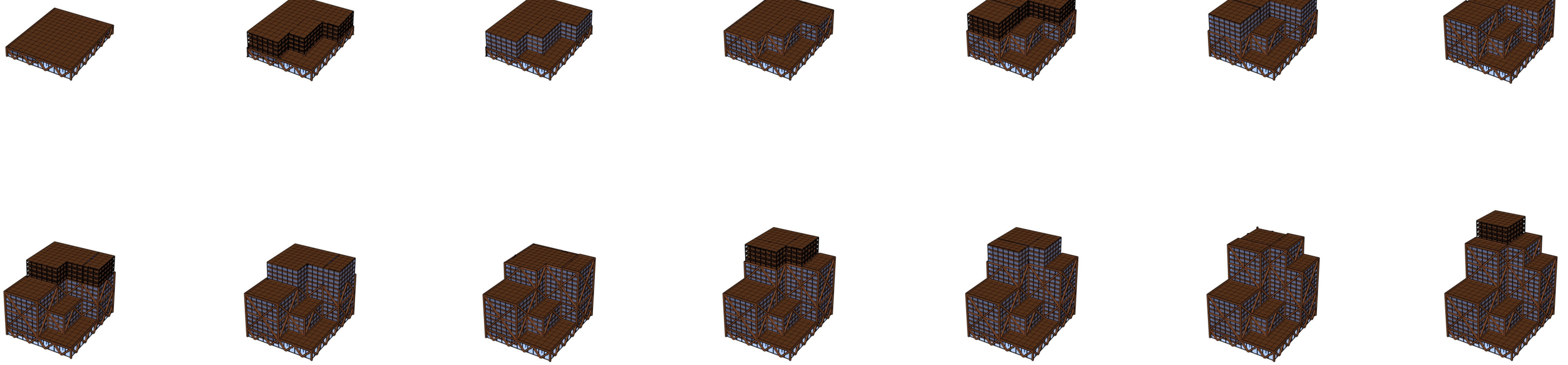
Stresses in top of tower (simplistic model)



Stresses in bottom elements (simplistic model)

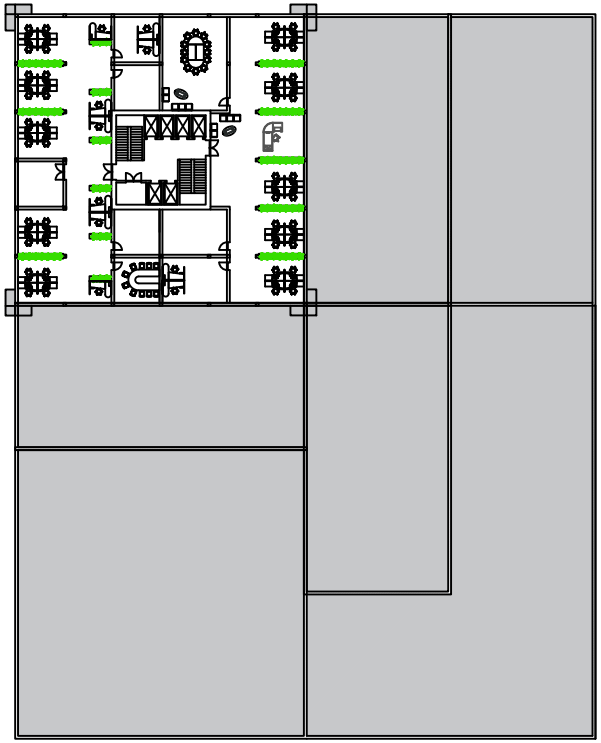


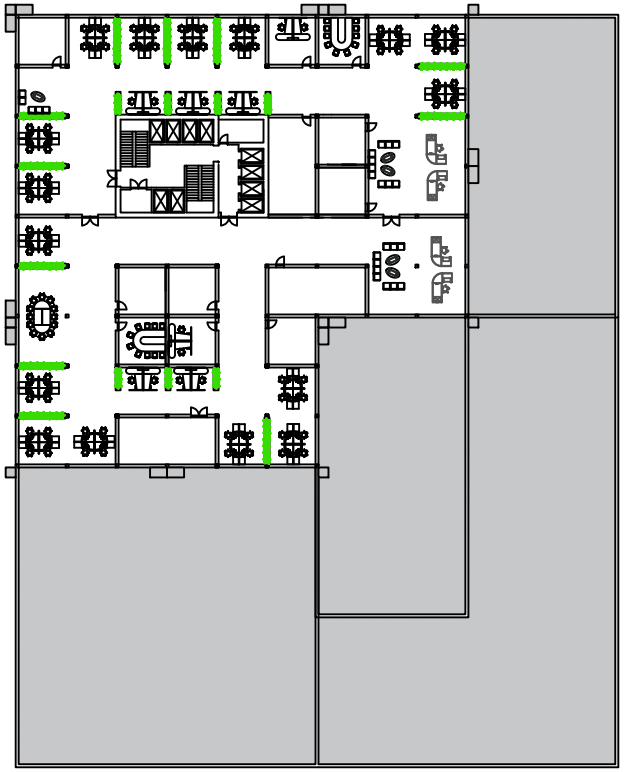
A.10 Construction plan

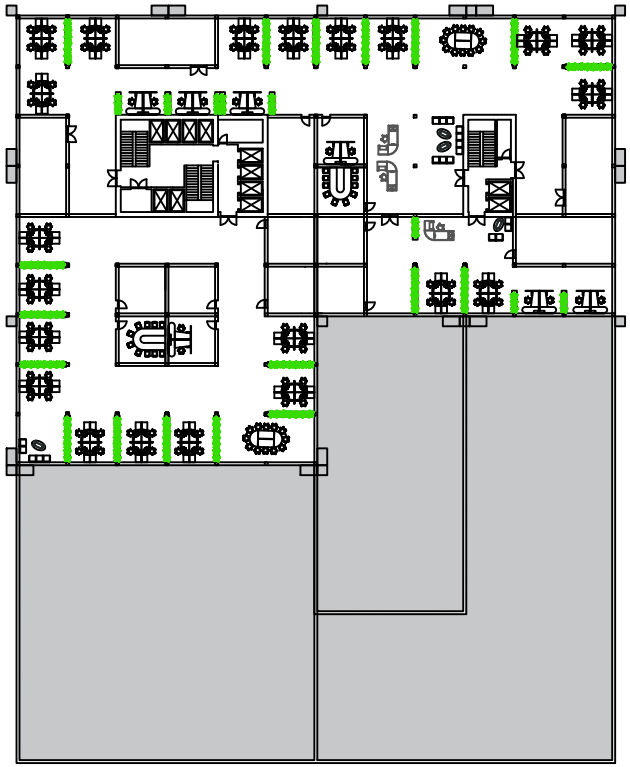


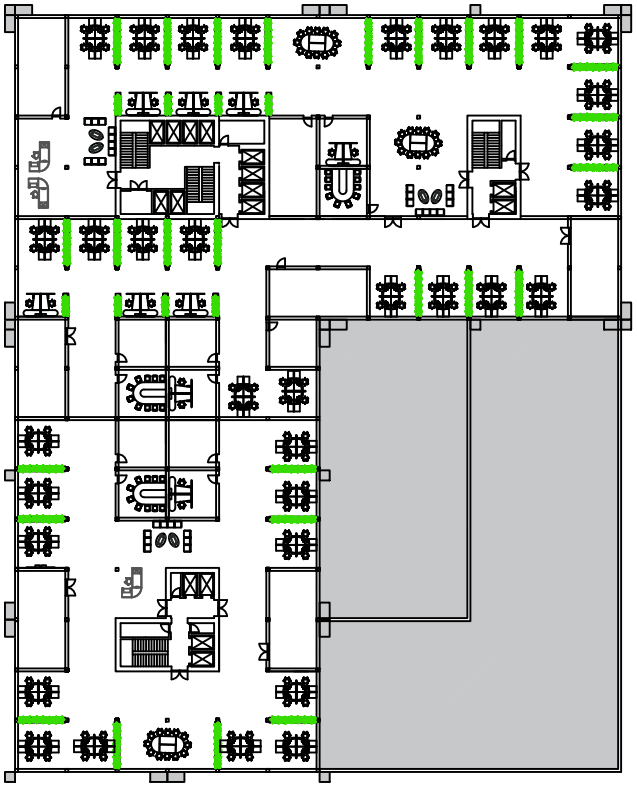
A.11 Physical model

A.12 Floorplans, sections, and elevations

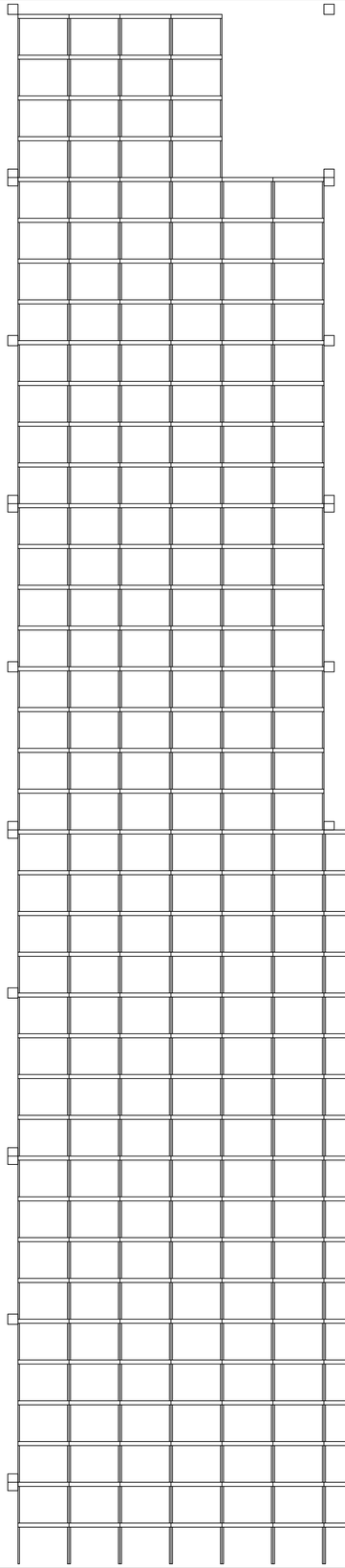


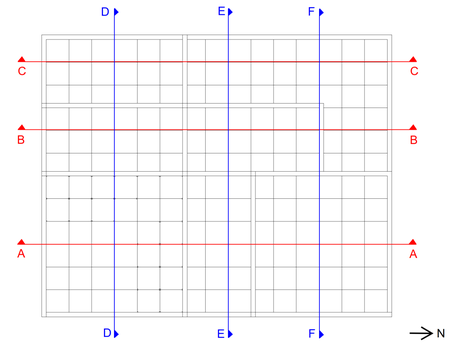




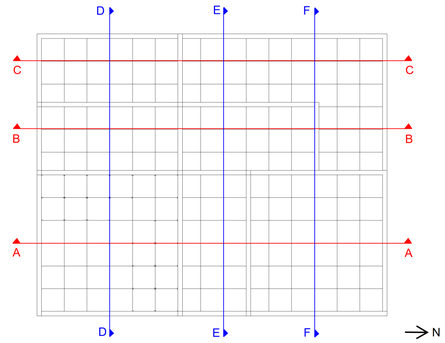


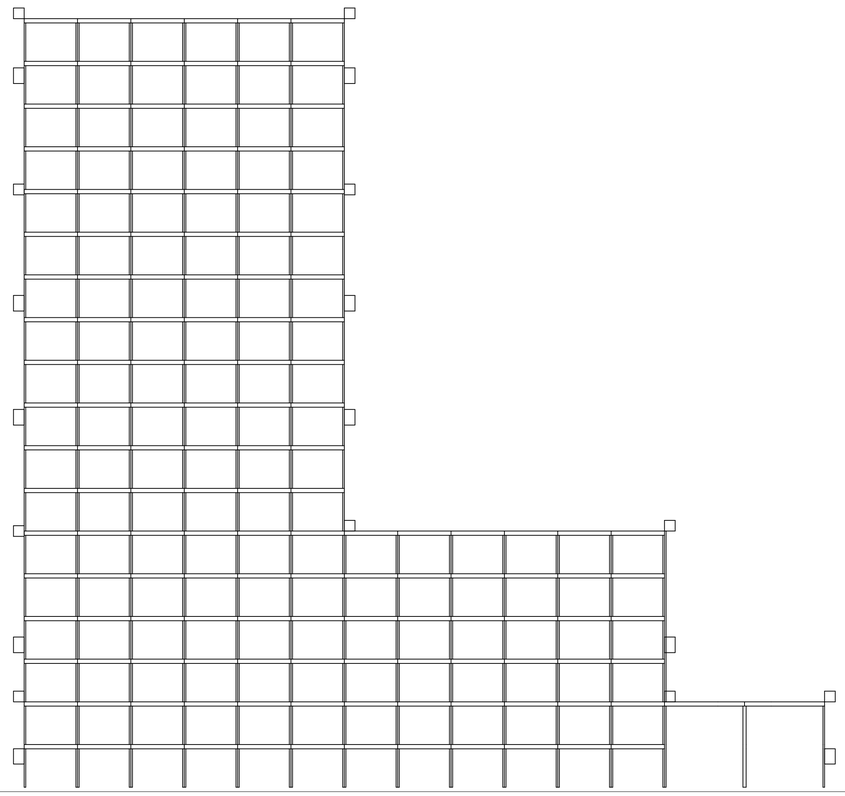
Section A



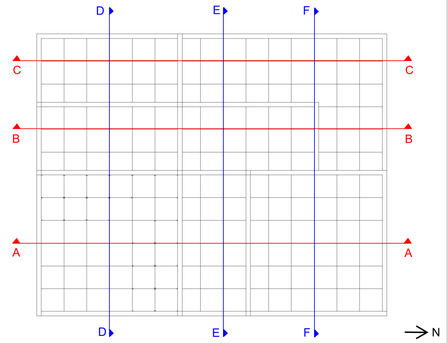


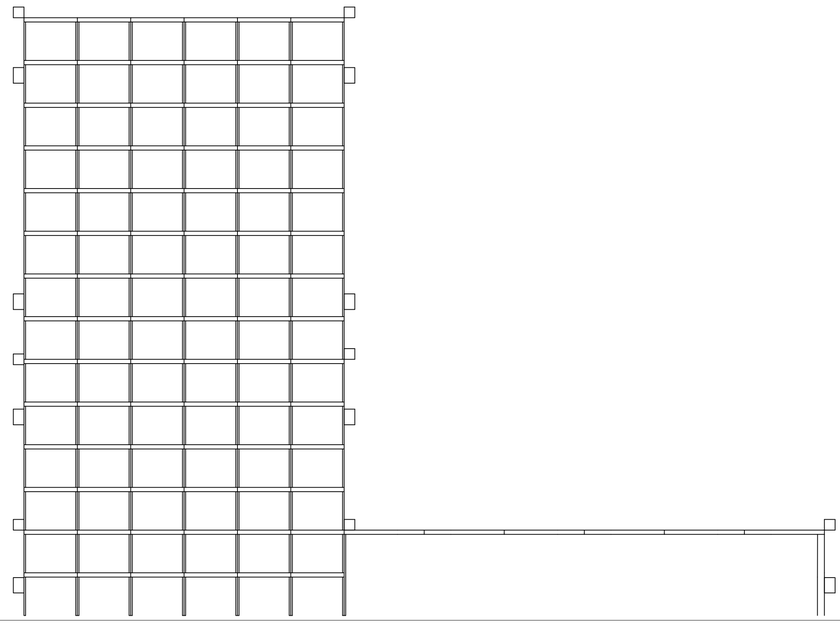
Section B



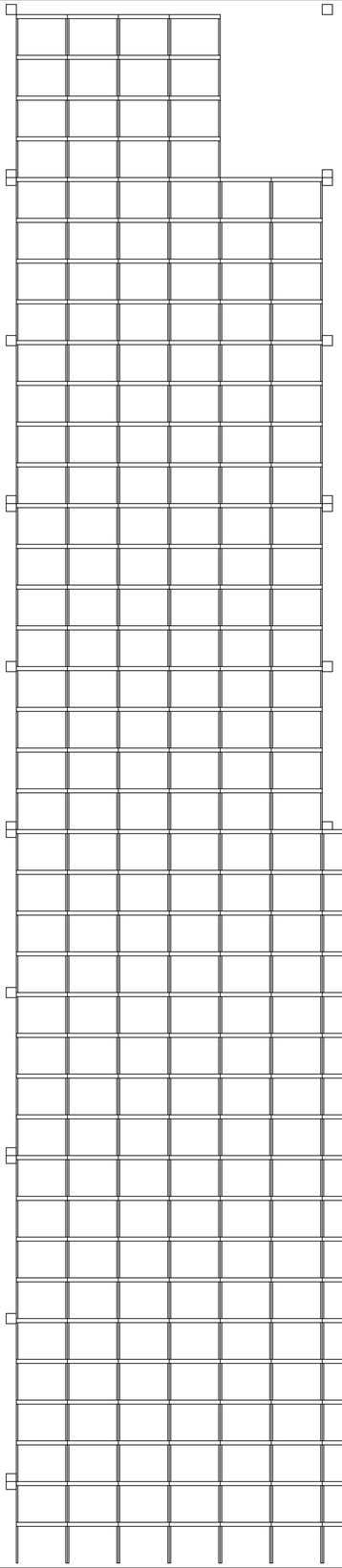


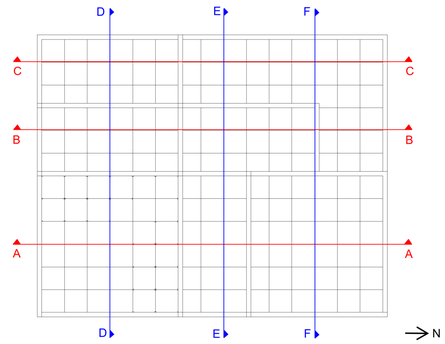
Section C



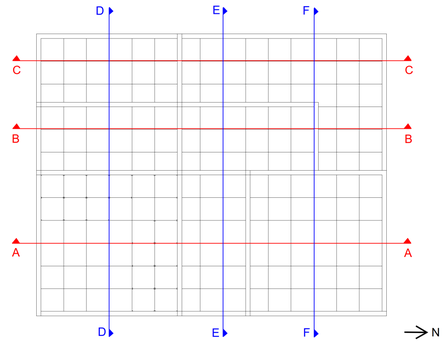


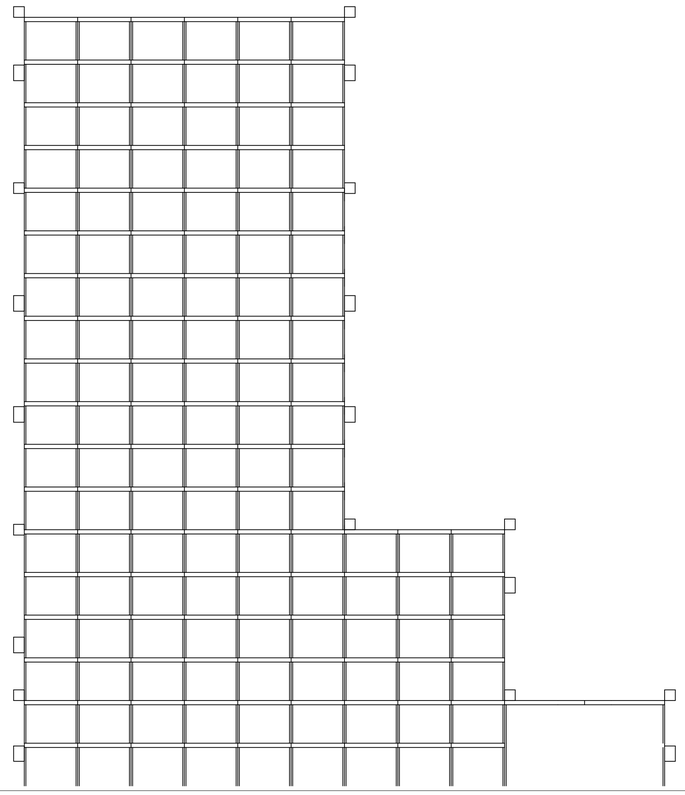
Section D



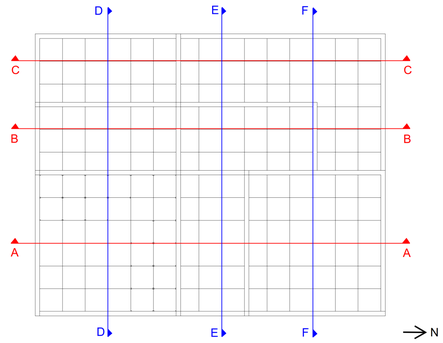


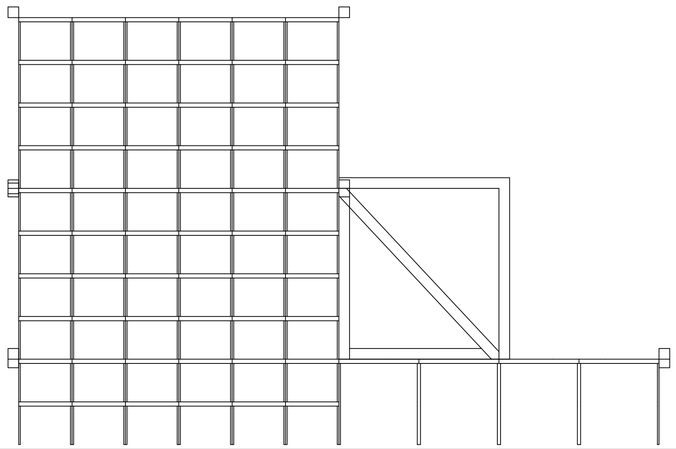
Section E

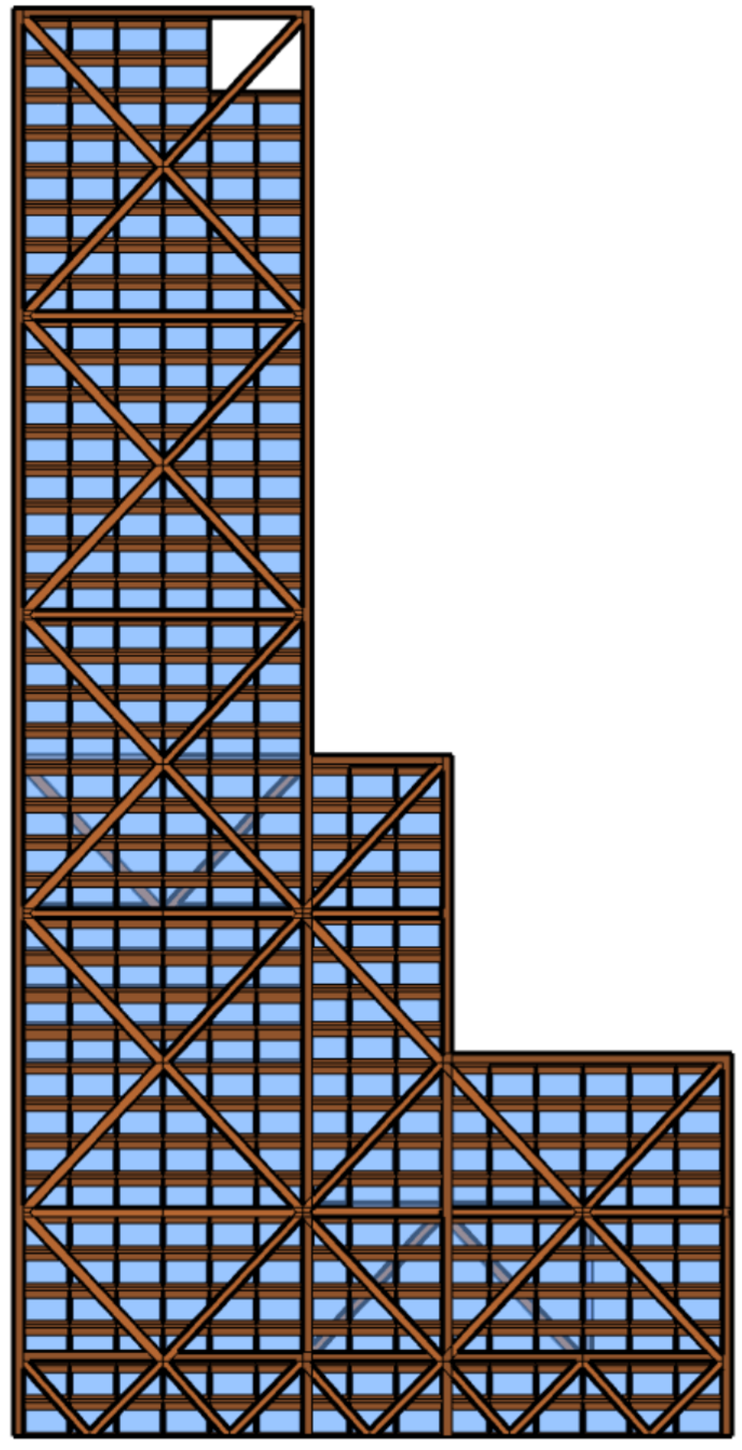




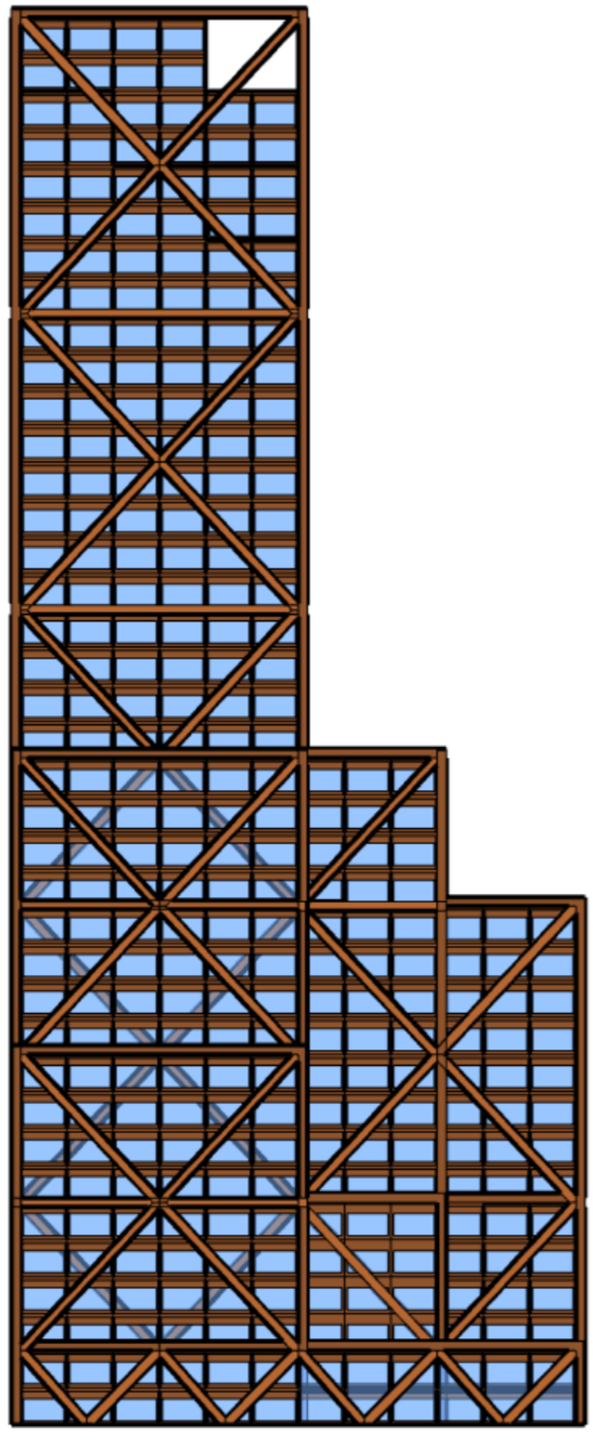
Section F



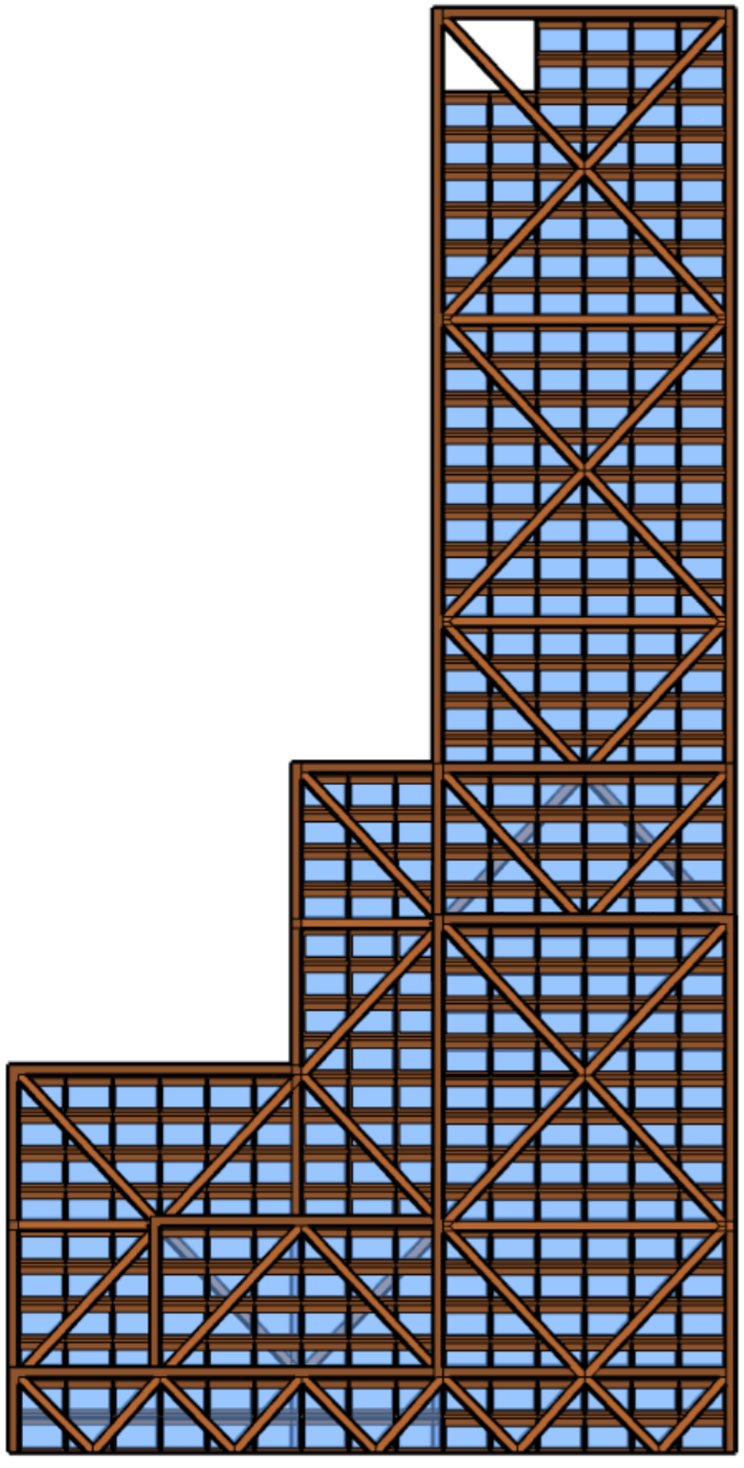




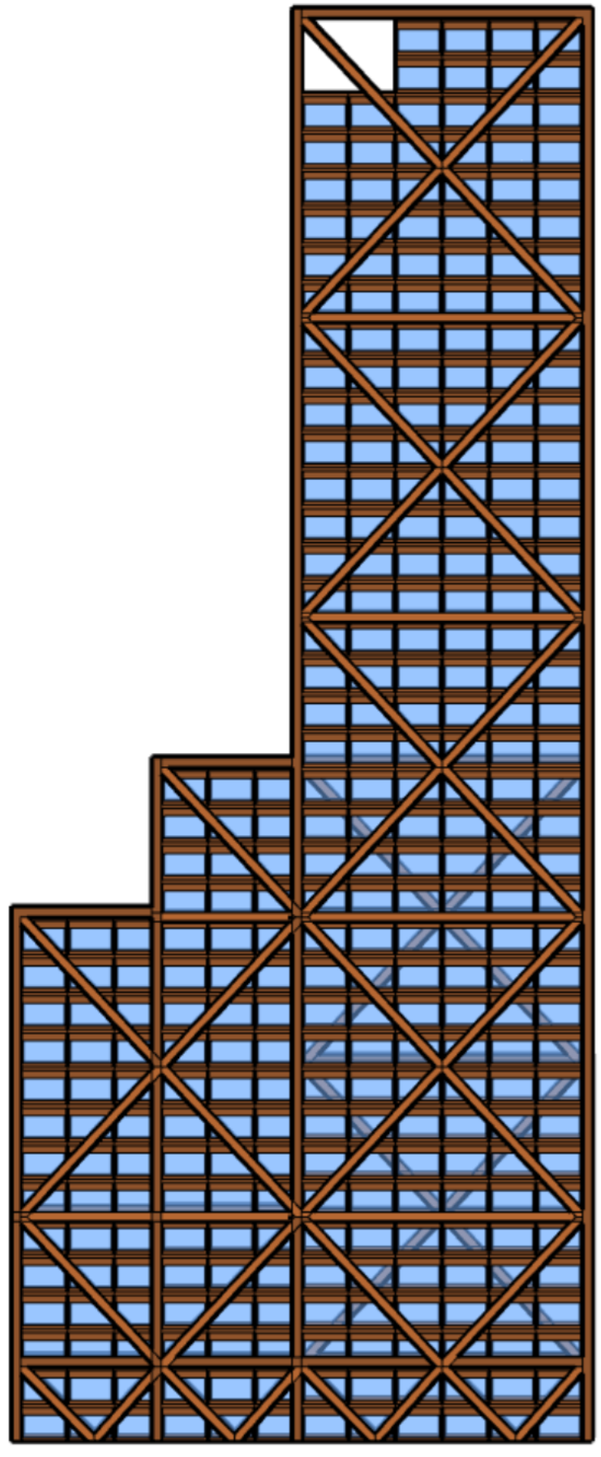
Elevations—front view



Elevations—right view

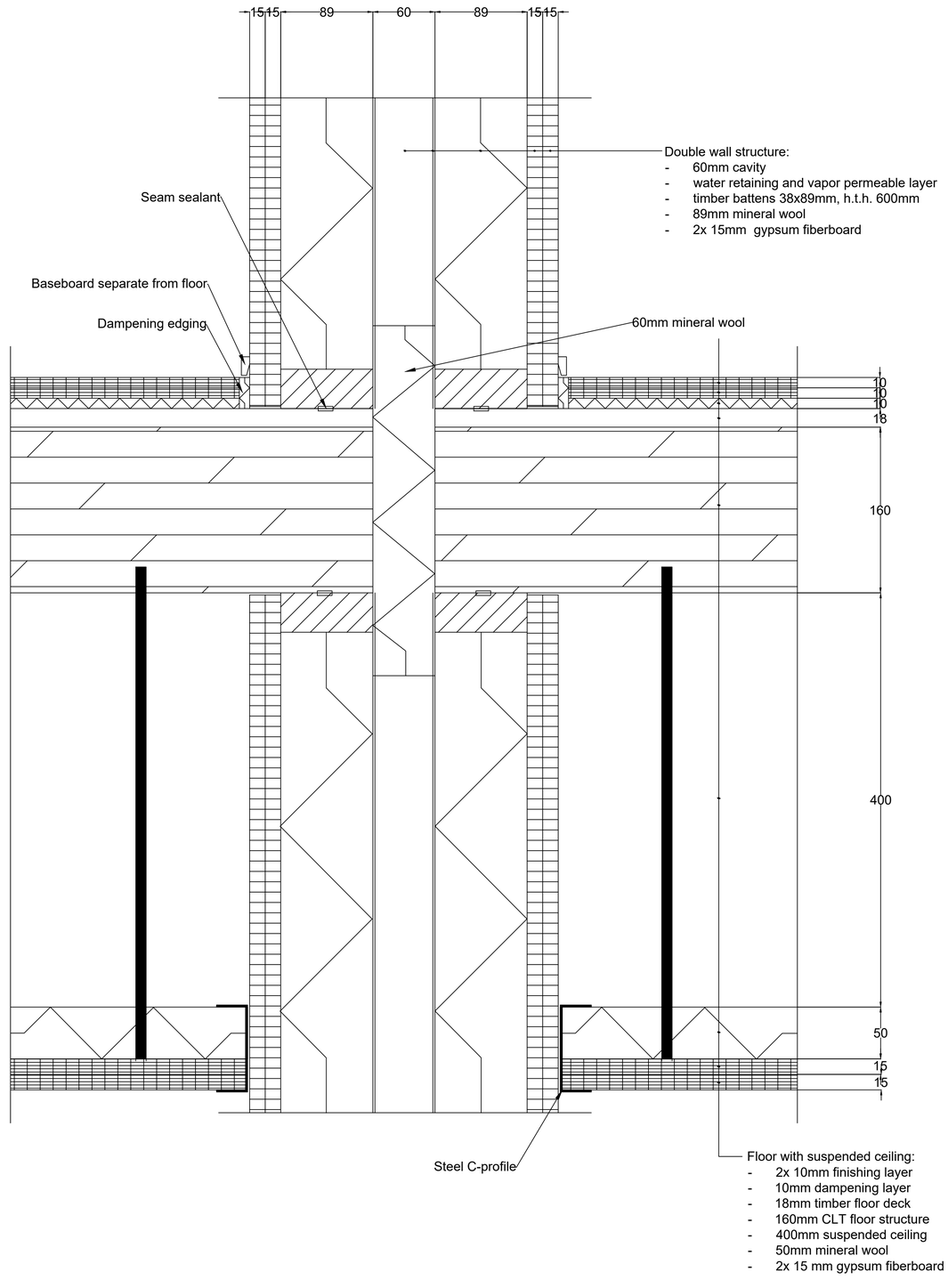


Elevations—back view



Elevations—left view

A.13 Acoustic remedial measures


15
15
89
60
89
15
15
Double wall structure:
- 60mm cavity
- water retaining and vapor permeable layer
- timber battens 38x89mm, h.t.h. 600mm
- 89mm mineral wool
- 2x 15mm gypsum fiberboard
Seam sealant
Baseboard separate from floor
Dampening edging
60mm mineral wool
10
10
10
18
160
400
50
15
15
Steel C-profile
Floor with suspended ceiling:
- 2x 10mm finishing layer
- 10mm dampening layer
- 18mm timber floor deck
- 160mm CLT floor structure
- 400mm suspended ceiling
- 50mm mineral wool
- 2x 15 mm gypsum fiberboard

A.14 Fire safety

Fire safety

Columns and beams

	Fire safety	R120			Effective Char depth		at 120 min	
	Duration	120	minutes		dchar	Bn*t		
B0	Charring rate	0,65-0,7	mm/min	Glulam			84	mm
Bn	effective char	0,7	mm/min		aeff	dchar+7		
	zero strength	7	mm				91	mm
				4 sides	so we reduce the actual by 2*91			

Here the difference between the UC between firesafty and regular can be found in the columns.

	Dimensioning inculding reduction due to fire safety						
	Verdiepingen		7	7	7	7	6
		Chosen dimensions		Chosen material		Gl32H	
ULS		Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 6
	witdh [mm]	720	720	720	640	600	520
	height [mm]	720	720	720	640	600	520
	area [mm^2]	518400	518400	518400	409600	360000	270400
	I = bh^3/12	22394880000	22394880000	22394880000	13981013333	10800000000	6093013333
	i = sqrt(I/A)	207,8460969	207,8460969	207,8460969	184,7520861	173,2050808	150,11107
	slenderness	0,812999791	0,812999791	0,812999791	0,914624765	0,97559975	1,125692019
	ϕ	0,88178431	0,88178431	0,88178431	0,979731707	1,043457411	1,216160463
	x	0,817517779	0,817517779	0,817517779	0,75135792	0,707411694	0,5964996
	Nrd	8984,585795	8136,983361	8136,983361	5908,919118	4889,629632	3096,835046
	UC-ULS	0,480477565	0,474682331	0,376953616	0,249932587	0,302033396	0,220100361
Fire safety SLS							
	reduced width [	538	538	538	458	418	338
	reduced height	538	538	538	458	418	338
	reduced Area [m	289444	289444	289444	209764	174724	114244
	I = bh^3/12	6981485761	6981485761	6981485761	3666744641	2544039681	1087640961
	i = sqrt(I/A)	155,3072224	155,3072224	155,3072224	132,2132116	120,6662063	97,57219549
	slenderness	1,088029461	1,088029461	1,088029461	1,278078275	1,400382416	1,731833875
	ϕ	1,170707	1,170707	1,170707	1,414549866	1,590573697	2,142807673
	x	0,623889059	0,623889059	0,623889059	0,494868062	0,426476033	0,2937115
	Nrd	3467,154139	3467,154139	3467,154139	1993,065678	1430,699488	644,2517114
	UC-SLS	0,881574536	0,788777217	0,626381907	0,807155063	0,730875697	0,749107697

And here for the beams, as we can see the bending stress is almost above the allowable stress of 19.2 MPa for GL24H

Fire ULS			Bending stress ULS			Bending stress SLS fire	
I = bh^3/12	1,186E+09	mm^4	sigma = M / W			sigma = M / W	
W = bh^2/6	5059103	mm^3	8,2552061	N/mm^2		18,9291369	N/mm^2

Finally for the cores we can see below what it adds (2 sides)

	The 10 x 10 core goes upto 152 meter						Fire safety						
	The core replaces 9 columns						R120						
							tfire [min]	120	min				
	CLT types	F,ck [MPa]	gamma	F,cd [MPa]	E-mod [MPa]		effective cha	0,65	mm/min				
	C24	21	1,3	16,1538462	11000		1 side	85	mm				
	C30	24	1,3	18,4615385	12000		2 side	170	mm				
	C35	25	1,3	19,2307692	13000								
	GL24H	24	1,3	18,4615385	11000								
	GL28H	28	1,3	21,5384615	12000								
	GL32H	32	1,3	24,6153846	13000								
For core A1					A1 dimensions								
	ULS	cumulative p	per 9 columns		Grid	10 x 10	meters						
	Zone 1	3056,5548	27508,9932	kN	Number of wa	4							
	Zone 2	2734,81219	24613,3097	kN	Wall length [r	10	m						
	Zone 3	2171,76262	19545,8636	kN	Wall length [r	10000	mm						
	Zone 4	1608,71305	14478,4175	kN	Total wall len	40000	mm						
	Zone 5	1045,66348	9410,97136	kN									
	Zone 6	482,613916	4343,52525	kN									
	Area required per zone			Thickness per zone		Adding firesafety		Final thickenss		Ply buildup			
	Zone 1	1702937,68	mm^2	42,5734419		212,573442	mm	220	7-ply	40	24 x 5	40	mm
	Zone 2	1523681,08	mm^3	38,092027		208,092027	mm	220	7-ply	40	24 x 5	40	mm
	Zone 3	1209982,03	mm^4	30,2495508		200,249551	mm	200	7-ply	40	24 x 5	40	mm
	Zone 4	896282,987	mm^5	22,4070747		192,407075	mm	200	7-ply	40	24 x 5	40	mm
	Zone 5	582583,942	mm^6	14,5645985		184,564599	mm	200	7-ply	40	20 x 5	40	mm
	Zone 6	268884,896	mm^7	6,7221224		176,722122	mm	180	7-ply	40	20 x 5	40	mm

A.15 Comparison CO_2 emission timber and steel trusses

Timber trusses + Concrete core

CLT Trusses

Grid size		
Width	5	m
Depth	5	m
Height story	4	m

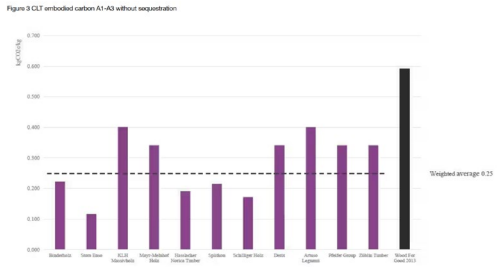
Tower 1		*Tower 2*		*Tower 3*		*Tower 4*		*Tower 5*	
Height	150 m	Height	80 m	Height	60 m	Height	40 m	Height	30 m
No. of stories	37	*No. of stories*	20	*No. of stories*	15	*No. of stories*	10	*No. of stories*	7
Dimensions		Dimensions		Dimensions		Dimensions		Dimensions	
Grid cells width	6	*Grid cells width*	4	*Grid cells width*	4	*Grid cells width*	3	*Grid cells width*	3
Grid cells depth	6	*Grid cells depth*	4	*Grid cells depth*	4	*Grid cells depth*	3	*Grid cells depth*	3

Dimensioning timber truss

Height	4	stories
Width	3	grid cells
Length truss	21.9317122	m

Dimensions truss

Width	1	m
Depth	1	m
Area truss	1	m2



Total length truss

Tower 1	1228.175883	m
Tower 2	241.2488342	m
Tower 3	131.5902732	m
Tower 4	87.7268488	m
Tower 5	43.8634244	m
Total volume trusses	1732.605264	m3
Embodied carbon CO2	0.25	kg CO2/kg
Density clt	490	kg/m3
Total weight	848976.5792	kg
Total embodied carbon trusses	**212244.1448**	**kg**

Concrete Cores

Main core		*Core 2*		*Core 3*		*Core 4*	
Height tower	150	Height tower	40 m	Height tower	50 m	Height tower	30 m
Dimensions		Dimensions		Dimensions		Dimensions	
Width	12 m	*Width*	8 m	*Width*	10 m	*Width*	6 m

Depth	12	m	*Depth*	8	m	*Depth*	10	m	*Depth*	6	m
Thickness wall	0.3	m	Thickness wall	0.3	m	Thickness wall	0.3	m	Thickness wall	0.3	m
Total area cross-section	14.04	m2	*Total area cross-section*	9.24	m2	*Total area cross-section*	11.64	m2	*Total area cross-section*	6.84	m2
Total volume core	2106	m3	*Total volume core*	369.6	m3	*Total volume core*	582	m3	*Total volume core*	205.2	m3

Amount of cores	1	
Total volume cores combined	2106	m3
Total weight cores	4843800	kg
Total embodied carbon concrete cores	**789539.4**	**kg**
TOTAL EMBODIED CARBON	1001783.545	kg CO2
Modification factor	0.24178	kg/kg
Embodied carbon concrete	**0.163**	**kg/kg**
Density concrete	2300	kg/m3

Steel trusses + CLT core

Steel Trusses

Grid size			
	Width	6	m
	Depth	6	m

Height story		4	m

Tower 1			*Tower 2*			*Tower 3*			*Tower 4*			*Tower 5*		
Height		150 m	Height		50 m	Height		30 m	Height		50 m	Height		40 m
	No. of stories	37		*No. of stories*	12		*No. of stories*	7		*No. of stories*	12		*No. of stories*	10
Dimensions			Dimensions			Dimensions			Dimensions			Dimensions		
	Grid cells width	5		*Grid cells width*	4		*Grid cells width*	3		*Grid cells width*	4		*Grid cells width*	3
	Grid cells depth	5		*Grid cells depth*	4		*Grid cells depth*	3		*Grid cells depth*	4		*Grid cells depth*	3

Dimensioning steel truss		
Height	2	stories
Width	2	grid cells
Length truss	14.4222051	m
Dimensions truss		
CHS diameter	250	mm
Thickness wall	16	mm
Area truss	0.011762123	m2

Total length truss		
Tower 1	1730.664612	m
Tower 2	403.8217429	m
Tower 3	86.53323061	m
Tower 4	374.9773326	m
Tower 5	144.222051	m
Total volume trusses	32.23079228	m3
Embodied carbon steel	1.94	kg CO2/kg
Density clt	7800	kg/m3
Total weight	251400.1798	kg
Total embodied carbon trusses	**487716.3487**	**kg**

CLT Cores

Main core			*Core 2*			*Core 3*			*Core 4*		
Height tower		150 m	Height tower		40 m	Height tower		50 m	Height tower		30 m
Dimensions			Dimensions			Dimensions			Dimensions		
	Width	12 m		*Width*	8 m		*Width*	10 m		*Width*	6 m
	Depth	12 m		*Depth*	8 m		*Depth*	10 m		*Depth*	6 m

Thickness wall	0.5	m	Thickness wall	0.5	m	Thickness wall	0.5	m	Thickness wall	0.5	m
Total area cross-section	23	m2	*Total area cross-section*	15	m2	*Total area cross-section*	19	m2	*Total area cross-section*	11	m2
Total volume core	3450	m3	*Total volume core*	600	m3	*Total volume core*	950	m3	*Total volume core*	330	m3

Amount of cores	3	
Total volume cores combined	5000	m3
Embodied carbon timber	0.25	kg CO2/kg
Density clt	490	kg/m3
Total weight	2450000	kg
Total embodied carbon clt cores	**612500**	**kg**
TOTAL EMBODIED CARBON	1100216.349	kg CO2

A.16 Calculation total CO_2 emission

Inventory materials

Members			Amount of	Material	w	h	l	m^3 per elem	total m^3	
Column	Zone									
		1	196	GL32H	0.498	0.498	4	0.992016	194.43514	
		2	343	GL32H	0.498	0.498	4	0.992016	340.26149	
		3	343	GL32H	0.458	0.458	4	0.839056	287.79621	
		4	553	GL32H	0.418	0.418	4	0.698896	386.48949	
		5	952	GL32H	0.378	0.378	4	0.571536	544.10227	
		6	1032	GL32H	0.338	0.338	4	0.456976	471.59923	
Beams										
Length		5	5746	GL32H	0.26	0.52	5	0.676	3884.296	
Diagonals										
Length		21.9317		GL32H	1	1	21.9317	21.9317122	0	
Cores				Thickness	Total lengt	Floor heigt	Floor amount			
Size		A1 - 1		0.8	40	4	38		4864	
		A1 - 2		0.8	20	4	18		1152	
		A2		0.8	15	4	6		288	
		A3		0.8					0	
		C4		0.8	30	4	14		1344	
		E5		0.8	40	4	10		1280	
Facade			Area [m^2	Thickness	Thickness [m]					27240
Glass			17978.4		0.012				215.7408	
Gypsum baord			9261.6		0.03				277.848	
Timber battens (20%)			9261.6		0.285				527.9112	
Mineral wool			9261.6		0.285				2111.6448	
Timber cladding			9261.6		0.03				277.848	
Floors										
Finishing			66600		0.04				2664	
Levelling layer			66600		0.02				1332	
Acoustic layer			66600		0.03				1998	
CLT floor panel			66600		0.16				10656	
Suspended ceiling			66600		0.015				999	
		mineral wool	66600		0.05				3330	
		gypsum	66600		0.0125				832.5	
Roofing (terrace)										
Paving finish			900		0.06				54	
Drainage layer			900		0.05				45	
Protection layer			900		0.005				4.5	
Waterproofing			900		0.005				4.5	
Thermal insulation			900		0.12				108	
Vapour barrier			900		0.0005				0.45	
CLT panel			900		0.14				126	
Ceilling			900		0.015				13.5	
Roofing (regular)										

Area: 66600 m2

Properties materials

Density:

GL32H	490	kg/m3
Glass	2490	
Gypsum board	800	
Timber battens	510	
Mineral wool	140	
Timber cladding	510	
CLT	490	
Tiles	1900	
Plastic	1380	
Concrete	2300	

Carbon emission

GL32H	0.42
Glass	0.91
Gypsum board	0.39
Timber battens	0.31
Mineral wool	1.28
Timber cladding	0.31
CLT	0.36
Tiles	0.48
Plastic	3.31
Concrete	0.40478

Carbon emission

Members			
Column	Zone		
		1	40014.75099
		2	70025.81423
		3	59228.45961
		4	79539.53663
		5	111976.2476
		6	97055.12195
Beams			
Length		5	799388.1168
Diagonals			
Length		21.9317122	0
Cores			
Size		A1 - 1	4528354.816
		A1 - 2	1072505.088
		A2	268126.272
		A3	0
		C4	1251255.936
		E5	1191672.32
Facade			
Glass			488847.0787
Gypsum baord			86688.576
Timber battens (20%)			83462.76072
Mineral wool			378406.7482
Timber cladding			43927.7688
Floors			
Finishing			421178.4
Levelling layer			
Acoustic layer			358041.6
CLT floor panel			1879718.4
Suspended ceiling			
		mineral wool	596736
		gypsum	259740
Roofing (terrace)			
Paving finish			49248
Drainage layer			
Protection layer			
Waterproofing			20555.1
Thermal insulation			19353.6
Vapour barrier			2055.51
CLT panel			22226.4
Ceilling			
Roofing (regular)			

Total 14279328.42 kg

214.4043307 kg/m2